

### 1. Features and Benefits

- Triaxis® Hall Technology
- 4 Triaxis® Magnetic Pixels
- > 7 Magnetic modes supported with off-chip DSP.
- 20-80mT Magnetic Range
- 3.3V Supply
- -40°C to 125°C or -40°C to 160°C Ambient Temperature Range
- Selectable Bus Protocol - Slave node
- Up to 10MHz SPI
- Up to 1MHz I2C
- Selectable 12-bits Digital Output
- Magnetic (Bx, By, Bz,  $\Delta B_x$ ,  $\Delta B_y$ ,  $\Delta B_z$ )
- Supply or Input Pin Voltage (V)
- Temperature (T)
- At runtime programmable modes
- Single Measurement mode
- Burst mode
- Wake-On-Change (WoC) mode
- On-chip magnetic calibration data allowing off-chip DSP to achieve maximum accuracy
- ISO26262 ASIL B Safety Element out of Context
- AEC-Q100 Qualified (Grade 0)
- Package(s) RoHS compliant

### 2. Application Examples

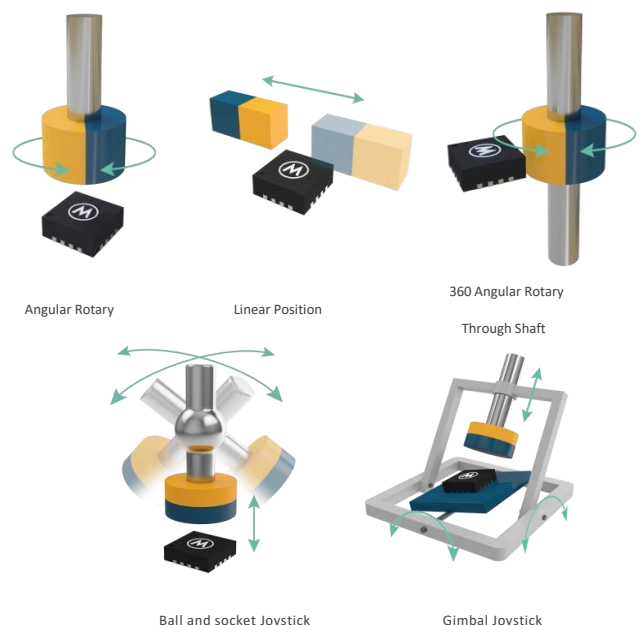
- Stalks
- Smart valves
- Shifters
- HMI
- Door handle
- Robotics

### 3. Description

The MLX90396 is a monolithic Magnetic Node sensor IC. It consists of 4 Triaxis® Hall magnetic front ends, an analog to digital conditioner, a DSP for temperature and offset compensation. The sensor is sensitive to the three components of the flux density applied to the IC (i.e. Bx, By and Bz), to the three differential magnetic fields ( $\Delta B_x$ ,  $\Delta B_y$  and  $\Delta B_z$ ), and to the temperature.

Its unique 4 magnetic pixels architecture enables the design of various non-contacting position sensors that are frequently required for both automotive and industrial applications.

The MLX90396 selectable measurement data can be read over I2C or SPI bus. The sensor is runtime programmable and offers on-chip temperature and offset compensation of the measured units.



## 4. Ordering Information

| Product  | Temp. | Package | Option Code | Packing Form | Definition                                  |
|----------|-------|---------|-------------|--------------|---------------------------------------------|
| MLX90396 | K     | LW      | AAA-000     | RE           | Rotary Linear Legacy and Stray field Robust |
| MLX90396 | G     | LW      | AAA-600     | RE           | Angular Rotary 360° Stray field Robust      |
| MLX90396 | G     | LW      | AAA-000     | RE           | Rotary Linear Legacy and Stray field Robust |
| MLX90396 | K     | LW      | BAA-000     | RE           | Rotary Linear Legacy and Stray field Robust |
| MLX90396 | G     | LW      | BAA-600     | RE           | Angular Rotary 360° Stray field Robust      |
| MLX90396 | G     | LW      | BAA-000     | RE           | Rotary Linear Legacy and Stray field Robust |

### Legend:

|                              |                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------|
| Temperature Code:            | <b>G:</b> from -40°C to 160°C<br><b>K:</b> from -40°C to 125°C                                               |
| Package Code:                | “LW” for QFN-16                                                                                              |
| Option Code - chip revision: | xxx-abc<br><b>Axx:</b> Engineering version<br><b>Bxx:</b> standard version                                   |
| Option Code - application:   | xxx- <b>abc</b> :<br><b>a:</b> Magnetic Mode<br>0: IMC Version<br>6: No IMC Version                          |
| Packing Form:                | “RE for Reel”                                                                                                |
| Ordering Example:            | “MLX90396GLW-BAA-600-RE”<br>For a Rotary Strayfield Robust application in QFN-16 package, delivered in Reel. |

*Table 1: Ordering information*

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## 5. General Description

The MLX90396 has 4 sensing spots, each capable of measuring the three vector components of the magnetic flux density ( $B_x$ ,  $B_y$  and  $B_z$ ) applied to the IC are sensed through the sensor front-end. The respective Hall signals ( $V_x$ ,  $V_y$  and  $V_z$ ) are generated at the Hall plates and amplified

The analog front-end is based on a fully differential analog chain featuring the classic offset cancellation technique (Hall plate 4 Phases spinning and chopper-stabilized amplification).

The conditioned analog signals are converted through a sigma-delta ADC (19 bits) and provided to a DSP block for further processing. The DSP stage is a custom finite state machine whose primary function consists of signal conditioning of the raw XYZ signals: offset & sensitivity adjustment and thermal offset & sensitivity drift compensation using the calibrated on-board temperature sensor. The result of these computations are a predetermined transfer characteristic of the sensor for a fixed gain setting.

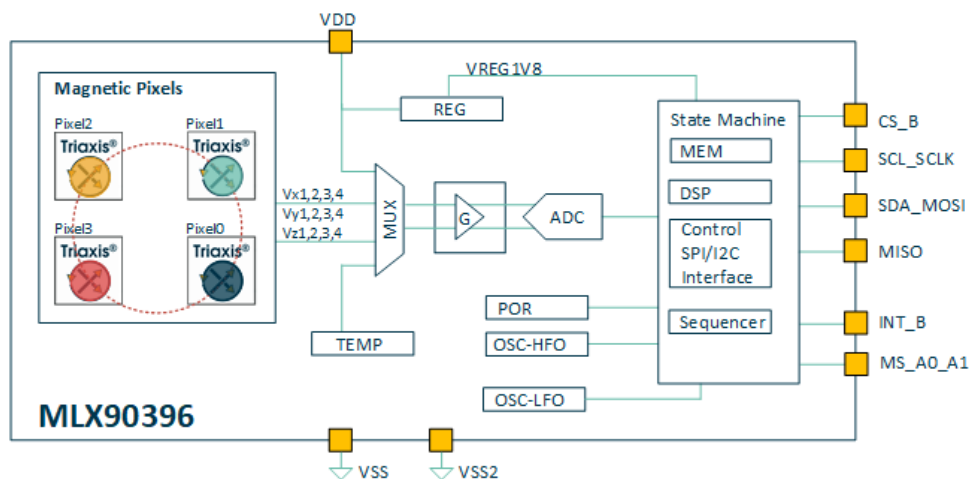


Figure 1 Block diagram for the QFN-16 package

These 3D linear Hall measurements are made available to the communication bus master for readout. The supported communication protocols are both I<sup>2</sup>C and SPI, where the Triaxis<sup>®</sup> Magnetic Node is a slave on the bus. Upon readback, it is then up to the microcontroller to process the measurement data – either in a vectorial or a scalar way – in order to achieve more complex position sensing functionality required by the application.

When processing data in a vectorial (or angular) way, all signal amplitude variations due to the sensitivity imperfections of the sensor, as well as the flux density degradation of the permanent magnet over temperature are cancelled out. The magnetic vectorial information is intrinsically self-compensated vs. flux density variations. This feature allows therefore an improved thermal accuracy vs. position sensors based on conventional 1D linear Hall sensors.

Another advantage of vectorial versus scalar signal processing is extended absolute position sensing range and higher linearity for Axial Parallel linear stroke position sensing applications.

In addition to the improved thermal accuracy, the realized position sensor features excellent linearity performances considering typical manufacturing tolerances (e.g. relative placement between the Hall IC and the magnet).

## 6. Glossary of Terms

| Term                 | Description                                       | Term | Description                      |
|----------------------|---------------------------------------------------|------|----------------------------------|
| Gauss (G), Tesla (T) | Units for the magnetic flux density:<br>1mT = 10G | EMC  | Electro-Magnetic Compatibility   |
| TC                   | Temperature Coefficient (in ppm/°C)               | FIR  | Finite Impulse Response          |
| ADC                  | Analog-to-Digital Converter                       | OSR  | Over Sampling Rate               |
| DAC                  | Digital-to-Analog Converter                       | PWL  | Piece Wise Linear                |
| LSB                  | Least Significant Bit                             | POR  | Power On Reset                   |
| MSB                  | Most Significant Bit                              | SW   | Software                         |
| DNL                  | Differential Non-Linearity                        | HW   | Hardware                         |
| INL                  | Integral Non-Linearity                            | IMC  | Integrated Magnetic Concentrator |
| ASP                  | Analog Signal Processing                          | ROM  | Read-only Memory                 |
| DSP                  | Digital Signal Processing                         | RAM  | Random-access Memory             |
| CDM                  | Charge Device Model                               | ESD  | Electrostatic Discharge          |
| HBM                  | Human Body Model                                  |      |                                  |

Table 2: Glossary of Terms

## 7. Pin Definitions

### 7.1. Pin Definition for QFN-16 Package

| Pin # | Name (I <sup>2</sup> C) | Name (SPI) | Description                                           |
|-------|-------------------------|------------|-------------------------------------------------------|
| 1     | A0_A1                   | MS         | [I] Mode Select / I2C address                         |
| 2     | NA                      | /CS        | [I] connect to GND / Chip Select                      |
| 3     | SCL                     | SCLK       | [I] Bus clock                                         |
| 4     | Not Used                | Not Used   | Not Connected                                         |
| 5     | SDA                     | MOSI       | [I/O] I <sup>2</sup> C Data / SPI Master Out Slave In |
| 6     | Not Used                | MISO       | [O] SPI Master In Slave Out                           |
| 7     | INT/TRG                 | INT/TRG    | [O or I] Interrupt / Trigger Pin (Programmable)       |
| 8     | Not Used                | Not Used   | Not Connected                                         |
| 9     | Not Used                | Not Used   | Not Connected                                         |
| 10    | Not Used                | Not Used   | Not Connected                                         |
| 11    | Not Used                | Not Used   | Not Connected                                         |
| 12    | Not Used                | Not Used   | Not Connected                                         |
| 13    | VSS                     | VSS        | [S] Ground                                            |
| 14    | VSS                     | VSS        | [S] Ground                                            |
| 15    | VDD                     | VDD        | [S] Supply                                            |
| 16    | Not Used                | Not Used   | Not Connected                                         |

Table 3: QFN-16 Pin Definitions and Descriptions

For optimal EMC behaviour connect the unused pins (Not Used) to the Ground (See section17). [I] stands for Input Pin, [O] stands for Output Pin and [S] stands for Supply Pin.

## 8. Absolute Maximum Ratings

| Parameter                  | Symbol                          | Min. | Typ. | Max. | Unit | Condition     |
|----------------------------|---------------------------------|------|------|------|------|---------------|
| Supply Voltage             | VDD <sub>MAX</sub>              |      |      | 4.0  | V    |               |
| Reverse Voltage Protection | VDD <sub>REV</sub>              | -0.3 |      |      | V    |               |
| Positive Output Voltage    | V <sub>out</sub>                |      |      | 4.0  | V    |               |
| Negative Output Voltage    | V <sub>out</sub> <sub>REV</sub> | -0.3 |      |      | V    |               |
| Positive Input Voltage     | V <sub>in</sub>                 |      |      | 4.0  | V    |               |
| Negative Input Voltage     | V <sub>in</sub> <sub>REV</sub>  | -0.3 |      |      | V    |               |
| Operating Temperature      | T <sub>AMB</sub>                | -40  |      | +125 | °C   | Option code K |
|                            |                                 | -40  |      | +160 |      | Option code G |
| Junction Temperature       | T <sub>J</sub>                  | -40  | 35   | +175 | °C   |               |
| Storage Temperature        | T <sub>ST</sub>                 | -55  |      | +150 | °C   |               |
| Magnetic Flux Density      | B <sub>MAX</sub>                | -1   |      | +1   | T    |               |
| ESD voltage, HBM           | ESD <sub>HBM</sub>              | -2   |      | 2    | kV   |               |
| ESD voltage, CDM           | ESD <sub>CDM</sub>              | -500 |      | 500  | V    |               |

*Table 4: Absolute Maximum Ratings*

Exceeding the absolute maximum ratings may cause permanent damage. Exposure to absolute maximum-rated conditions for extended periods may affect device reliability.

## 9. General Electrical Specification

General electrical specifications are valid for temperature range: -40 - 160°C, supply voltage range: 3.0 - 3.6V unless otherwise noted (condition).

| Parameter                | Symbol                 | Min. | Typ. | Max. | Unit | Condition                                   |
|--------------------------|------------------------|------|------|------|------|---------------------------------------------|
| Analog Supply Voltage    | VDD                    | 3    | 3.3  | 3.6  | V    |                                             |
| Supply Current           | I <sub>dd.conv1p</sub> |      | 4    | 6.5  | mA   | 1 pixel measurement                         |
| Supply Current           | I <sub>dd.conv2p</sub> |      | 5    | 7.5  | mA   | 2 pixel measurement                         |
| Supply Current           | I <sub>dd.convT</sub>  |      | 4    | 6.5  | mA   | Temperature measurement                     |
| Supply Current           | I <sub>dd.convV</sub>  |      | 4    | 6.5  | mA   | Voltage measurement                         |
| Ready Current            | I <sub>dd.ready</sub>  |      | 2.8  | 3.5  | mA   | All Temp                                    |
| Idle Current             | I <sub>dd.idle</sub>   |      | 2.5  | 4.0  |      | T <sub>A</sub> = -40°C                      |
|                          |                        |      | 3.5  | 4    |      | T <sub>A</sub> = room temp                  |
|                          |                        |      | 5    | 8    | μA   | T <sub>A</sub> = 85°C                       |
|                          |                        |      | 10   | 20   |      | T <sub>A</sub> = 125°C                      |
|                          |                        |      | 30   | 75   |      | T <sub>A</sub> = 160°C                      |
| Sleep Current            | I <sub>dd.slp</sub>    |      | 40   | 65   |      | T <sub>A</sub> = -40°C                      |
|                          |                        |      | 75   | 140  | nA   | T <sub>A</sub> = room temp                  |
|                          |                        |      | 140  | 500  |      | T <sub>A</sub> = 85°C                       |
|                          |                        |      | 0.6  | 3.3  | μA   | T <sub>A</sub> = 125°C                      |
|                          |                        |      | 4    | 23   |      | T <sub>A</sub> = 160°C                      |
| Power-On reset (rising)  | V <sub>porLH</sub>     | 1.36 | 1.46 | 1.46 | V    |                                             |
| Power-On reset (falling) | V <sub>porLH</sub>     | 1.32 | 1.41 | 1.6  | V    |                                             |
| Input High Threshold     | V <sub>IH</sub>        |      |      | 70   | %VDD | SDA/MOSI, SCL/SCLK                          |
| Input Low Threshold      | V <sub>IL</sub>        | 30   |      |      | %VDD | SDA/MOSI, SCL/SCLK                          |
| Input Level Hysteresis   | V <sub>IHYST</sub>     | 0.15 | 0.74 | 1.1  | %VDD | SDA/MOSI, SCL/SCLK                          |
| Input High Threshold     | V <sub>IH</sub>        |      |      | 70   | %VDD | INT/TRG, MS/CS                              |
| Input Low Threshold      | V <sub>IL</sub>        | 30   |      |      | %VDD | INT/TRG, MS/CS                              |
| Input Level Hysteresis   | V <sub>IHYST</sub>     | 0.15 | 0.74 | 1.1  | %VDD | INT/TRG, MS/CS                              |
| Output Level High        | V <sub>OH</sub>        | 90   |      |      | %VDD | MISO, INT, INT/TRIG, SDA (Static, 1mA load) |
| Output Level Low         | V <sub>OL</sub>        |      |      | 10   | %VDD | MISO, INT, INT/TRIG, SDA (Static, 1mA load) |
| Leakage Current          | I <sub>leak</sub>      |      |      | 1    |      | SDA                                         |
|                          |                        |      |      | 1    |      | INT                                         |
|                          |                        |      |      | 1    |      | MISO                                        |
|                          |                        |      |      | 0.8  | μA   | SCL_SCLK                                    |
|                          |                        |      |      | 0.8  |      | MS_A0_A1                                    |
|                          |                        |      |      | 0.8  |      | MOSI                                        |

Table 5: Electrical Specification

## 10. Timing Specification

Timing conditions include the variations of supply, temperature and aging, unless specified.

| Parameter                                                | Symbol                 | Min. | Typ. | Max. | Unit                | Condition                                                                                        |
|----------------------------------------------------------|------------------------|------|------|------|---------------------|--------------------------------------------------------------------------------------------------|
| Main Oscillator Frequency                                | F <sub>OSC</sub>       |      | 4    |      | MHz                 | T <sub>A</sub> = 35°C                                                                            |
| Main Oscillator Thermal Drift                            | F <sub>OSC_TH</sub>    | -4   |      | 4    | %F <sub>OSC</sub>   | T <sub>A</sub> within T <sub>AMB</sub>                                                           |
| Low Power Oscillator Frequency                           | F <sub>LPOSC</sub>     |      | 50   |      | kHz                 | T <sub>A</sub> = 35°C                                                                            |
| LP Oscillator Thermal Drift                              | F <sub>LPOSC_TH</sub>  | -25  |      | 20   | %F <sub>LPOSC</sub> | T <sub>A</sub> within T <sub>AMB</sub>                                                           |
| Reset Time (POR and RT command)                          | T <sub>POR</sub>       |      | 0.6  | 1.2  | ms                  | Time from STARTUP to READY                                                                       |
| Temperature Measurement and Conversion Time <sup>1</sup> | T <sub>TEMP</sub>      |      | 19   |      | μs                  | Preceding each measurement sequence<br>F <sub>OSC</sub> = 4MHz                                   |
| Magnetic Component Measurement and Conversion Time       | T <sub>MAG</sub>       |      | 77   |      | μs                  | For each magnetic component measurement<br>F <sub>OSC</sub> = 4MHz                               |
| Voltage Measurement and Conversion Time                  | T <sub>VDD</sub>       |      | 14   |      | μs                  | VDD measurement<br>F <sub>OSC</sub> = 4MHz                                                       |
| Input Pin Measurement and Conversion Time                | T <sub>INPUT</sub>     |      | 14   |      | μs                  | VDD measurement<br>F <sub>OSC</sub> = 4MHz                                                       |
| Residual processing time after measurement(s)            | T <sub>RESIDUAL</sub>  |      | 16.5 |      | μs                  | Valid only if ADC diagnostic is not on<br>F <sub>OSC</sub> = 4MHz                                |
| ADC Diagnostic Processing Time                           | T <sub>ADCDIAG</sub>   |      | 44   |      | μs                  | Valid only if ADC diagnostic is on (T <sub>RESIDUAL</sub> is skipped)<br>F <sub>OSC</sub> = 4MHz |
| Trigger Pulse Width                                      | T <sub>TRIG</sub>      | TBD  |      | TBD  | μs                  |                                                                                                  |
| Memory Store time                                        | T <sub>HS_WAIT</sub>   |      | TBD  |      | ms                  | T <sub>A</sub> within T <sub>AMB</sub>                                                           |
| T Sleep to READY mode                                    | T <sub>slp2ready</sub> |      | 0.6  | 1.2  | ms                  | All temperature, all Vdd                                                                         |
| T Startup to READY mode                                  | T <sub>str2ready</sub> |      | 0.6  | 1.2  | ms                  | All temperature, all Vdd, from supply to INT_B rising edges                                      |
| Burst rate fast mode                                     | BR_F                   | 0.0  |      | 31.5 | ms                  | BURST_DATA_RATE_STEP = 0 (0.5ms/LSB) F <sub>LPOSC</sub> = 50kHz                                  |
| Burst rate slow mode                                     | BR_S                   | 0.0  |      | 1260 | ms                  | BURST_DATA_RATE_STEP = 1 (20ms/LSB) F <sub>LPOSC</sub> = 50kHz                                   |

Table 6: General Timing Specification

<sup>1</sup> Examples of total conversion time calculation:

Temperature measurement + 2 \* Magnetic measurement + VDD or Input Pin measurement:

$$T_{TEMP} + 2 * T_{MAG} + T_{VDD} \text{ OR } T_{INPUT} + T_{RESIDUAL} = 19 + 2 * 77 + 14 + 16.5 = 203.5 \mu s$$

Temperature measurement + 6 \* Magnetic measurement + ADC diagnostic:

$$T_{TEMP} + 6 * T_{MAG} + T_{ADCDIAG} = 19 + 6 * 77 + 44 = 525 \mu s$$

## 11. Temperature Specification

The MLX90396 allows for measuring of the junction temperature for on-chip thermal compensations as well as off-chip thermal information within the application, provided there is proper modelling of the junction temperature to application temperature.

| Parameter                      | Symbol       | Min. | Typ. | Max. | Unit                 | Condition                   |
|--------------------------------|--------------|------|------|------|----------------------|-----------------------------|
| Temperature Sensor Resolution  | $T_{RES}$    |      | 1    |      | LSB <sub>9</sub> /°C |                             |
| Temperature Sensor Output 0°C  | $T_0$        |      | 0    |      | LSB <sub>9</sub>     |                             |
| Temperature Sensor Output 35°C | $T_{35}$     |      | 35   |      | LSB <sub>9</sub>     | Calibration Temperature     |
| Temperature Sensor Accuracy    | $T_{LINERR}$ | -10  |      | 10   | °C                   | $T_A \leq 35^\circ\text{C}$ |
| Temperature Sensor Accuracy    | $T_{LINERR}$ | -8   |      | 8    | °C                   | $T_A > 35^\circ\text{C}$    |

Table 7: Temperature Specification

## 12. Voltage Specification

The MLX90396 allows for measurement of the supply voltage for ADC plausibility checks as well as for monitoring battery discharge in battery powered applications.

| Parameter               | Symbol    | Min. | Typ. | Max. | Unit                 | Condition |
|-------------------------|-----------|------|------|------|----------------------|-----------|
| Voltage Resolution      | $V_{RES}$ |      | TBD  |      | LSB <sub>12</sub> /V |           |
| Voltage Output for 3.3V | $V_{3V}$  |      | 1100 |      | LSB <sub>12</sub>    |           |
| Voltage Accuracy        | $V_{ACC}$ | TBD  |      | TBD  | mV                   |           |

Table 8: Voltage Specification

## 13. Magnetic Field Requirements

This section describes the magnetic field requirements to meet the performance described in section 14.

### 13.1. IMC Variant (-0xx code)

| Parameter                       | Symbol     | Min. | Typ. | Max. | Unit | Condition                  |
|---------------------------------|------------|------|------|------|------|----------------------------|
| Magnetic Flux Density in X or Y | $B_X, B_Y$ |      |      | 80   | mT   | $\sqrt{B_X^2 + B_Y^2}$     |
| Magnetic Flux Density in Z      | $B_Z$      |      |      | 170  | mT   |                            |
| k factor                        | k          | 1.06 | 1.08 | 1.09 |      | $G_{IMC\_XY} / G_{IMC\_Z}$ |

Table 9: Magnetic field requirement for IMC Variant

### 13.2. No IMC Variant (-6xx code)

| Parameter                  | Symbol | Min. | Typ. | Max. | Unit | Condition |
|----------------------------|--------|------|------|------|------|-----------|
| Magnetic Flux Density in Z | $B_Z$  |      |      | TBD  | mT   |           |

Table 10: Magnetic field requirement for NO IMC Variant

## 14. Accuracy Specification

### 14.1. IMC Variant (-0xx code)

Magnetic Specifications (with **Melexis default factory programming**) are valid for temperature range: -40 - 160°C, supply voltage range: 3.0 - 3.6V with **60mT for 1 pixel mode for pixel 0 and 1, or 10mT/mm for 2 pixel mode** unless otherwise noted (condition).

| Parameter                      | Symbol                                           | Min. | Typ.     | Max. | Unit                                          | Condition                              |
|--------------------------------|--------------------------------------------------|------|----------|------|-----------------------------------------------|----------------------------------------|
| Angular accuracy error         | NLE                                              | -1.2 |          | +1.2 | deg                                           | Derived value                          |
| Angular accuracy thermal drift | NLE                                              | -1   |          | +1   | deg                                           | Derived value                          |
| ADC External Resolution        | N <sub>ADCEXT</sub>                              |      | 12       |      | bits                                          |                                        |
| Sensing Range                  | B <sub>RANGE50</sub>                             | 20   | 60       | 80   | mT                                            | Limited performance for 10mT           |
| Offset                         | X <sub>0</sub> , Y <sub>0</sub> , Z <sub>0</sub> | -5   |          | +5   | LSB <sub>12</sub>                             | T <sub>A</sub> within T <sub>AMB</sub> |
| Offset drift                   |                                                  | -5   |          | +5   | LSB <sub>12</sub>                             |                                        |
| Sensitivity                    |                                                  |      | 50<br>20 |      | μT/LSB <sub>12</sub><br>LSB <sub>12</sub> /mT |                                        |
| Sensitivity Accuracy           |                                                  | -10  |          | +10  | %                                             | T <sub>A</sub> = 35°C                  |
| Sensitivity drift              |                                                  | -10  |          | 10   | %                                             | T <sub>A</sub> within T <sub>AMB</sub> |
| SMISM between axis             |                                                  | -3   |          | +3   | %                                             | X, Y and Z for PX0 and PX1             |
| SMISM pixel-to-pixel           |                                                  | -4   |          | 4    | %                                             | PX0 to other PX                        |
| Orthogonality Error            |                                                  | -0.5 |          | 0.5  | %                                             | X to Y, PX0 and PX1                    |
| Noise RMS                      |                                                  |      |          | 0.6  | LSB <sub>12</sub>                             | 1sigma                                 |
| Hysteresis                     |                                                  | -8   |          | +8   | LSB <sub>12</sub>                             | T <sub>A</sub> within T <sub>AMB</sub> |

Table 11: Accuracy Specification for IMC Variant 1 pixel mode

| Parameter                      | Symbol                                              | Min. | Typ.       | Max. | Unit                                                | Condition                                                                            |
|--------------------------------|-----------------------------------------------------|------|------------|------|-----------------------------------------------------|--------------------------------------------------------------------------------------|
| Angular accuracy error         | NLE                                                 | -2.4 |            | +2.4 | deg                                                 | Derived value                                                                        |
| Angular accuracy thermal drift | NLE                                                 | -2   |            | +2   | deg                                                 | Derived value                                                                        |
| ADC External Resolution        | N <sub>ADCEXT</sub>                                 |      | 12         |      | bits                                                |                                                                                      |
| Sensing Range                  | B <sub>RANGE50</sub>                                |      | 10         |      | mT/mm                                               |                                                                                      |
| Offset                         | X <sub>0</sub> , Y <sub>0</sub> ,<br>Z <sub>0</sub> | -25  |            | +25  | LSB <sub>12</sub>                                   | T <sub>A</sub> within T <sub>AMB</sub>                                               |
| Offset drift                   |                                                     | -15  |            | +15  | LSB <sub>12</sub>                                   |                                                                                      |
| Sensitivity                    |                                                     |      | 8.4<br>120 |      | μT/mm/LSB <sub>12</sub><br>LSB <sub>12</sub> /mT/mm |                                                                                      |
| Sensitivity Accuracy           |                                                     | -10  |            | +10  | %                                                   | T <sub>A</sub> = 35°C                                                                |
| Sensitivity drift              | TCSXY                                               | -10  |            | 10   | %                                                   | T <sub>A</sub> within T <sub>AMB</sub>                                               |
| SMISM between axis             |                                                     | -5   |            | +5   | %                                                   | T <sub>A</sub> within T <sub>AMB</sub>                                               |
| Orthogonality Error            |                                                     | -1.0 |            | 1.0  | %                                                   | X <sub>PX02</sub> to Z <sub>PX02</sub> and<br>Y <sub>PX13</sub> to Z <sub>PX13</sub> |
| Noise RMS                      |                                                     |      |            | 3.2  | LSB <sub>12</sub>                                   | 1sigma                                                                               |
| Hysteresis                     |                                                     |      | TBD        |      | LSB <sub>12</sub>                                   | T <sub>A</sub> within T <sub>AMB</sub>                                               |

Table 12: Accuracy Specification for IMC Variant 2 pixel mode

## 14.2. IMC Variant (-0xx code) low temperature

Magnetic Specifications (with **Melexis default factory programming**) are valid for temperature range: -40 - 125°C, supply voltage range: 3.0 - 3.6V with 60mT for 1 pixel mode for all pixel, or 10mT/mm for 2 pixel mode unless otherwise noted (condition).

| Parameter                      | Symbol                                           | Min. | Typ.     | Max. | Unit                                          | Condition                              |
|--------------------------------|--------------------------------------------------|------|----------|------|-----------------------------------------------|----------------------------------------|
| Angular accuracy error         | NLE                                              | -1.2 |          | +1.2 | deg                                           | Derived value                          |
| Angular accuracy thermal drift | NLE                                              | -1   |          | +1   | deg                                           | Derived value                          |
| ADC External Resolution        | N <sub>ADCEXT</sub>                              |      | 12       |      | bits                                          |                                        |
| Sensing Range                  | B <sub>RANGE50</sub>                             | 20   | 60       | 80   | mT                                            | Limited performance for 10mT           |
| Offset                         | X <sub>0</sub> , Y <sub>0</sub> , Z <sub>0</sub> | -5   |          | +5   | LSB <sub>12</sub>                             | T <sub>A</sub> within T <sub>AMB</sub> |
| Offset drift                   |                                                  | -5   |          | +5   | LSB <sub>12</sub>                             |                                        |
| Sensitivity                    |                                                  |      | 50<br>20 |      | μT/LSB <sub>12</sub><br>LSB <sub>12</sub> /mT |                                        |
| Sensitivity Accuracy           |                                                  | -10  |          | +10  | %                                             | T <sub>A</sub> = 35°C                  |
| Sensitivity drift              |                                                  | -10  |          | 10   | %                                             | T <sub>A</sub> within T <sub>AMB</sub> |
| SMISM between axis             |                                                  | -3   |          | +3   | %                                             | X, Y and Z for PX0 and PX1             |
| SMISM pixel-to-pixel           |                                                  | -4   |          | 4    | %                                             | PX0 to other PX                        |
| Orthogonality Error            |                                                  | -0.5 |          | 0.5  | %                                             | X to Y, PX0 and PX1                    |
| Noise RMS                      |                                                  |      |          | 0.6  | LSB <sub>12</sub>                             | 1sigma                                 |
| Hysteresis                     |                                                  | -8   |          | +8   | LSB <sub>12</sub>                             | T <sub>A</sub> within T <sub>AMB</sub> |

Table 13 : Accuracy Specification for IMC Variant 1 pixel mode low temperature

| Parameter                      | Symbol                                              | Min. | Typ.       | Max. | Unit                                                | Condition                                |
|--------------------------------|-----------------------------------------------------|------|------------|------|-----------------------------------------------------|------------------------------------------|
| Angular accuracy error         | NLE                                                 | -2.4 |            | +2.4 | deg                                                 | Derived value                            |
| Angular accuracy thermal drift | NLE                                                 | -2   |            | +2   | deg                                                 | Derived value                            |
| ADC External Resolution        | N <sub>ADCEXT</sub>                                 |      | 12         |      | bits                                                |                                          |
| Sensing Range                  | B <sub>RANGE50</sub>                                |      | 10         |      | mT/mm                                               |                                          |
| Offset                         | X <sub>0</sub> , Y <sub>0</sub> ,<br>Z <sub>0</sub> | -25  |            | +25  | LSB <sub>12</sub>                                   | T <sub>A</sub> within T <sub>AMB</sub>   |
| Offset drift                   |                                                     | -15  |            | +15  | LSB <sub>12</sub>                                   |                                          |
| Sensitivity                    |                                                     |      | 8.4<br>120 |      | μT/mm/LSB <sub>12</sub><br>LSB <sub>12</sub> /mT/mm |                                          |
| Sensitivity Accuracy           |                                                     | -10  |            | +10  | %                                                   | T <sub>A</sub> = 35°C                    |
| Sensitivity drift              | TCSXY                                               | -10  |            | 10   | %                                                   | T <sub>A</sub> within T <sub>AMB</sub>   |
| SMISM between axis             |                                                     | -5   |            | +5   | %                                                   | T <sub>A</sub> within T <sub>AMB</sub>   |
| Orthogonality Error            |                                                     | -1.0 |            | 1.0  | %                                                   | X_PX02 to Z_PX02 and<br>Y_PX13 to Z_PX13 |
| Noise RMS                      |                                                     |      |            | 3.2  | LSB <sub>12</sub>                                   | 1sigma                                   |
| Hysteresis                     |                                                     |      | TBD        |      | LSB <sub>12</sub>                                   | T <sub>A</sub> within T <sub>AMB</sub>   |

Table 14 : Accuracy Specification for IMC Variant 2 pixel mode low temperature

### 14.3. NO IMC Variant (-6xx code)

Magnetic Specifications (with **Melexis default factory programming**) are valid for temperature range: -40 - 160°C, supply voltage range: 3.0 - 3.6V with **30mT/mm** unless otherwise noted (condition).

| Parameter                    | Symbol                                           | Min. | Typ. | Max. | Unit                    | Condition |
|------------------------------|--------------------------------------------------|------|------|------|-------------------------|-----------|
| angular accuracy error       | NLE                                              |      | TBD  |      | deg                     |           |
| angular accuracy error drift | NLE                                              |      | TBD  |      | deg                     |           |
| ADC External Resolution      | N <sub>ADCEXT</sub>                              |      | TBD  |      | bits                    |           |
| Sensing Range                | B <sub>RANGE50</sub>                             |      | 30   |      | mT/mm                   |           |
| Offset                       | X <sub>0</sub> , Y <sub>0</sub> , Z <sub>0</sub> |      | TBD  |      | LSB <sub>12</sub>       |           |
| Offset drift                 |                                                  |      | TBD  |      | LSB <sub>12</sub>       |           |
| Sensitivity                  |                                                  |      | TBD  |      | μT/mm/LSB <sub>12</sub> |           |
| Sensitivity Accuracy         |                                                  |      | TBD  |      | %                       |           |
| Sensitivity thermal drift    |                                                  |      | TBD  |      | %                       |           |
| Noise                        |                                                  |      | TBD  |      | LSB <sub>12</sub>       |           |
| Hysteresis                   |                                                  |      | TBD  |      | deg                     |           |

**Table 15 : Accuracy Specification for NO-IMC Variant**

## 15. Functional Description

### 15.1. Operating Modes

The MLX90396 always acts as slave on the bus. It operates within the following 6 modes:

- Measurement mode
- Ready mode
- Idle mode
- Sleep mode
- Memory Store mode
- Communication mode

The six operating modes are controlled and defined by the master. All modes can be enabled and disabled at runtime through the communication protocol. On reset, the device is in ready mode, waiting for commands.

The components (field in X, Y or Z direction, temperature and voltage) which are needed to be measured can be provided when sending the commands. The order of conversion is fixed, and components which are not needed are simply skipped.

It is the user's responsibility to read back the data and ensure that timings are met. The MLX90396 is purely a slave. Even in burst mode, the user is responsible to collect the data from the MLX90396.

| Measurement Modes            | Start of Mode | End of Mode (Return to Ready*) | Measurement Data         |
|------------------------------|---------------|--------------------------------|--------------------------|
| Single Measurement           | SM command    | EX command                     | Argument of SM command   |
| Burst Mode                   | SB command    | EX command                     | Argument of SB command   |
| Wake-up On Change (WOC) Mode | SWOC command  | EX command                     | Argument of SWOC command |

*Table 16 : Measurement modes*

#### 15.1.1. Single Measurement

In single measurement mode, a single measurement is requested by the master through the SM command. A measurement can be repeated by toggling the CS\_B input.

#### 15.1.2. Burst Mode

In burst mode, the MLX90396 will continuously make measurements. The mode is started with the SB command and can be ended with the EX command. Measurements are read by the RM command and the status byte will indicate the freshness of the data. The rate of measurements is programmable. Each time a measurement is finished, the INT pin will go low. Reading out the data will clear the INT pin. If the measurement is not read on time, it will be overwritten by the next measurement when ready. The INT pin will show again a rising edge when the set of measurements can be read out. As long as the INT pin is high, the data is ensured to be most recent and ready.

### 15.1.3. Wake-up on Change Mode

In wake-up on change mode, the MLX90396 will continuously make measurements. The mode is started with the SWOC command and can be ended with the EX command. Measurements can be read by the RM command, the status byte will indicate the freshness of the data. Also here the rate of measurements is programmable. This mode differs from the burst mode in the sense that the INT pin will not go low at each finish of measurements but goes low only when a specific condition is met. This condition can be controlled by the user:

- Absolute mode: each measurement is compared with the initial one (first measurement after start of the WOC mode). If the difference is larger than the threshold, then the INT pin goes low. The pin is cleared by the RM command.
- Differential mode: each measurement is compared with the previous one. If the difference is larger than the threshold, then the INT pin goes low. The pin is cleared by the RM command.

### 15.2. Flow Chart

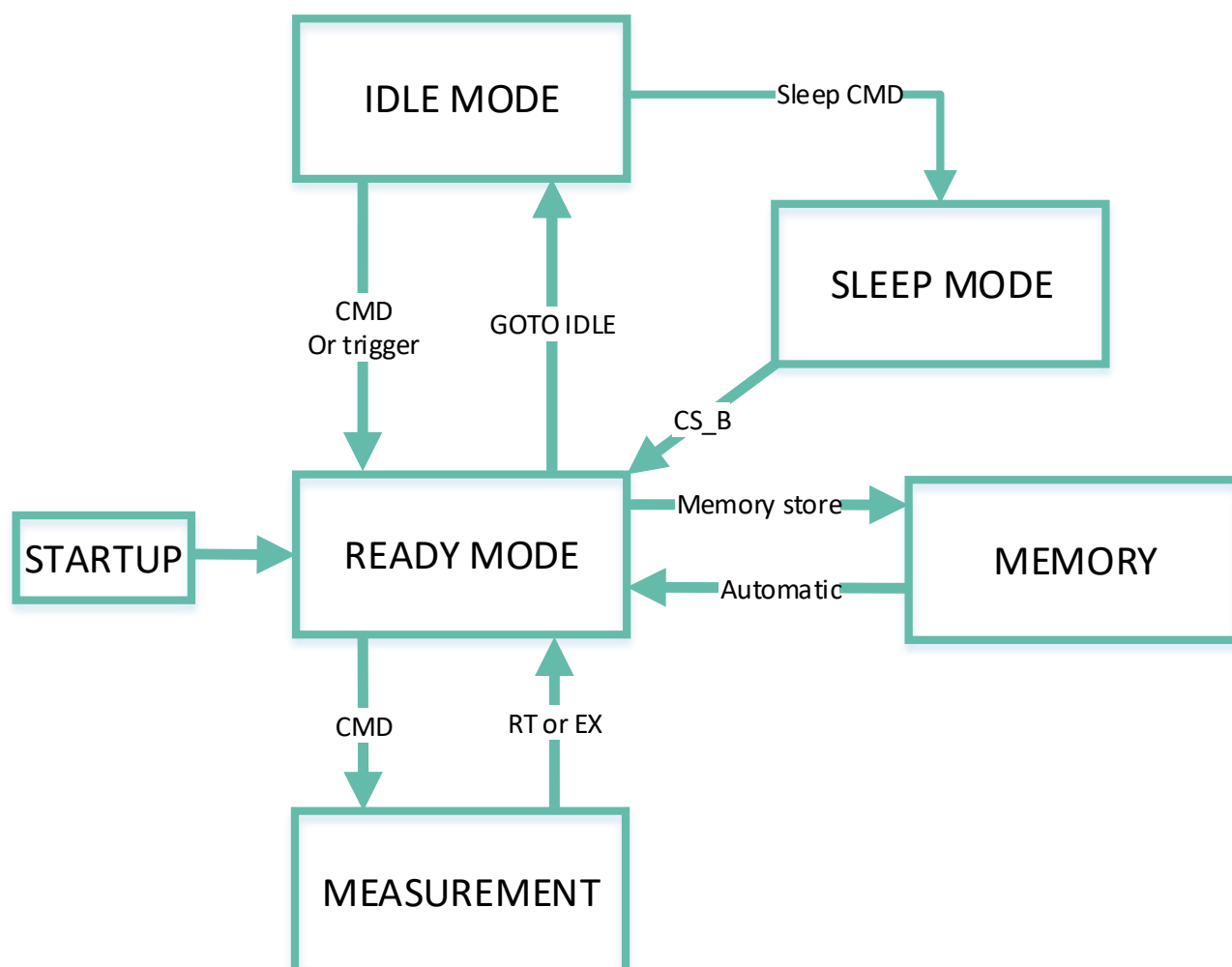


Figure 2: Flow chart

## 15.3. Status Byte

A status byte is provided on each command. Depending on the operating mode, the interpretation differs.

| Mode       | Function ID [7:6] |   | Measurement counter [5:3] |   |   | Flags [2:0] |           |      |
|------------|-------------------|---|---------------------------|---|---|-------------|-----------|------|
| Burst Mode | 1                 | 0 | Measurement Counter       |   |   | ERROR       | INFO/WARN | DRDY |
| WOC Mode   | 0                 | 1 | Measurement Counter       |   |   | ERROR       | INFO/WARN | DRDY |
| SM Mode    | 1                 | 1 | 0                         | 0 | 0 | ERROR       | INFO/WARN | DRDY |
| Exit       | 0                 | 0 | 0                         | 0 | 0 | ERROR       | INFO/WARN | DRDY |
| Memory     | 0                 | 0 | 0                         | 0 | 0 | ERROR       | INFO/WARN | DRDY |

Table 17: Status Byte

The SB command is always replied with a “Burst Mode” function ID, the SWOC command by a “WOC Mode” function ID. When the SM command is successfully received, an “SM Mode” function ID is given.

For the RM and RV command, the status returned purely depends on the mode in which the IC is operating. For single measurement mode, if the measurement is finished, the “Idle” status byte is given.

An EX command, when successfully received is replied with only the flags in the status byte.

Memory commands (HR, HS, RR and WR) when successfully received is replied with only the flags in the status byte.

In case of an illegal command (HR in burst mode for example) then the status will set the ERROR flag of the mode currently in, not necessarily the one associated with the command. The command got rejected, and after reading the ERROR register the CE\_INVALID bit is set.

Melexis recommends verifying that the status byte matches the expected value.

### 15.3.1. Data Ready- DRDY

This flag is set to 1 to indicate that new data is ready (not yet read), or when it is the first time that this data is being read. This flag is updated after the measurement is done.

### 15.3.2. Info/Warn

The Info/Warn flag is set whenever one bit of the WARNING registers is set. One can read the warning register at address 0x42 to get detailed information and clearing the warning flag. The result reading the warning register is as follows:

| Info/warning register [7:0] |       |       |        |     |     |     |     |
|-----------------------------|-------|-------|--------|-----|-----|-----|-----|
| UVTEMP                      | OVVDD | UVVDD | OVVDDD | SEC | OVF | RST | MI6 |

Table 18: info/warning register first byte

| Info/warning register [15:8] |       |       |       |       |       |       |        |
|------------------------------|-------|-------|-------|-------|-------|-------|--------|
| WOC_T                        | WOCM6 | WOCM5 | WOCM4 | WOCM3 | WOCM2 | WOCM1 | OVTEMP |

Table 19 : info/warning register second byte

| Flag Name | description                                   | Condition                                      |
|-----------|-----------------------------------------------|------------------------------------------------|
| WOC_T     | Wake up on change triggered by temperature    | After measurement is performed                 |
| WOC_M6    | Wake up on change triggered by measurement #6 | After measurement is performed                 |
| WOC_M5    | Wake up on change triggered by measurement #5 | After measurement is performed                 |
| WOC_M4    | Wake up on change triggered by measurement #4 | After measurement is performed                 |
| WOC_M3    | Wake up on change triggered by measurement #3 | After measurement is performed                 |
| WOC_M2    | Wake up on change triggered by measurement #2 | After measurement is performed                 |
| WOC_M1    | Wake up on change triggered by measurement #1 | After measurement is performed                 |
| OV_TEMP   | Over temperature                              | After measurement is performed                 |
| UV_TEMP   | Under temperature                             | After measurement is performed                 |
| OV_VDD    | Overvoltage on VDD                            | When event occurs in READY or Measurement mode |
| UV_VDD    | Under voltage on VDD                          | When event occurs in READY or Measurement mode |
| OV_VDDD   | Overvoltage on digital supply                 | When event occurs                              |
| SEC       | Memory single error correction                | During memory recall                           |
| OVF       | ADC overflow                                  | After measurement is performed                 |
| RST       | Reset flag                                    | At start up or after RST command               |
| MI6       | More than 6 measurement requested             | After SB, SWOC or SM command                   |

Table 20: Description of warning flags

### 15.3.3. ERROR

The ERROR flag is set whenever one bit of the ERROR registers is set. One can read the error register at address 0x41 to get detailed information and clearing the error flag. The result reading the error register is as follows:

| Error register [7:0] |          |          |           |        |     |            |        |
|----------------------|----------|----------|-----------|--------|-----|------------|--------|
| DIAG_OSC             | DIAG_LFO | DIAG_ADC | BUS_ERROR | CRCERR | DED | CE_INVALID | CE_CRC |

Table 21: Error register first byte

| Error register [15:8] |   |   |   |   |           |         |          |
|-----------------------|---|---|---|---|-----------|---------|----------|
| -                     | - | - | - | - | DIAG_INTB | DIAG_HP | DIAG_DIE |

Table 22: Error register second byte

| Flag Name  | description                                               | Condition                                            |
|------------|-----------------------------------------------------------|------------------------------------------------------|
| DIAG_INTB  | Error in the INT_B read back                              | When the event occurs                                |
| DIAG_HP    | Error in hall element or gain chain detected              | After a measurement is performed                     |
| DIAG_DIE   | Die ring crack                                            | When the event occurs                                |
| DIAG_OSC   | Error in oscillators detected                             | When event occurs while in READY or MEASUREMENT mode |
| DIAG_LFO   | Error in low frequency oscillator detected                | When the event occurs                                |
| DIAG_ADC   | Error in ADC detected                                     | After measurement is performed                       |
| BUS_ERORR  | Error in internal bus                                     | When the event occurs                                |
| CRCERR     | Port CRC error                                            | When event occurs while in READY or MEASUREMENT mode |
| DED        | Memory double error detection                             | During memory recall                                 |
| CE_INVALID | Invalid command or command occurring during memory access | During communication                                 |
| CE_CRC     | Wrong command CRC error                                   | During communication                                 |

Table 23: description of error flags

### 15.3.4. Measurement counter

The 3-bit measurement counter is present in burst mode and WOC mode it is increased on each measurement. This allows the user to know if a measurement is missed when for example the RM commands are not in sync with the measurement rate.

### 15.3.5. Function ID

The function ID is shows which is the current measurement mode the chip is in.

- Burst mode: 0x02
- WOC mode: 0x01
- single measurement mode: 0x03

## 15.4. Commands

The following table shows a list of available commands

| Name                          | Symbol | # | CMD Byte #1                | CMD Byte #2 | CMD Byte #3 | CMD Byte #4 |
|-------------------------------|--------|---|----------------------------|-------------|-------------|-------------|
| Start Burst Mode              | SB     | 1 | 0b0001 DATA[3:0]           | DATA[7:0]   | DATA[7:0]   | CRC[7:0]    |
| Start WOC Mode                | SWOC   | 2 | 0b0010 DATA[3:0]           | DATA[7:0]   | DATA[7:0]   | CRC[7:0]    |
| Start Single Measurement Mode | SM     | 3 | 0b0011 DATA[3:0]           | DATA[7:0]   | DATA[7:0]   | CRC[7:0]    |
| Read Measurement              | RM     | 4 | 0b0100 T000                | CRC[7:0]    | N/A         | N/A         |
| Read Register                 | RR     | 5 | 0b0101 dddd <sup>(2)</sup> | REG[7:0]    | CRC[7:0]    | N/A         |
| Write Register                | WR     | 6 | 0b0110 dddd <sup>(2)</sup> | REG[7:0]    | DATA[7:0]   | DATA[7:0]   |
| Exit Mode                     | EX     | 8 | 0b1000 DATA[3:0]           | CRC[7:0]    | N/A         | N/A         |
| Memory Recall                 | HR     | D | 0b1101 dddd <sup>(2)</sup> | CRC[7:0]    | N/A         | N/A         |
| Memory Store                  | HS     | E | 0b1110 dddd <sup>(2)</sup> | CRC[7:0]    | N/A         | N/A         |
| Reset                         | RT     | F | 0b1111 0000                | CRC[7:0]    | N/A         | N/A         |

*Table 24: List of commands*

<sup>2</sup> d stands for “don’t care”. These bits are not checked and can be both 0 and 1.

### 15.4.1. Measurement commands

The MLX90396 has 3 measurement commands available: Burst Mode, WOC Mode and single measurement mode. Each command contains ‘function bits’ followed by the measurement data and an optional CRC if enabled.

The function bits are the first 4 bits that have to be send:

- Burst Mode it is 0x1
- WOC Mode it is 0x2
- Single Measurement it is 0x3

The next bits are the data bits for which measurements should be performed. The data bits are limited to a maximum of 6 magnetic measurements + Temperature and VDD.

| CMD Byte #1 |    |    |    |         |       |       |       | CMD Byte #2 |       |       |       |       |       |       |       |
|-------------|----|----|----|---------|-------|-------|-------|-------------|-------|-------|-------|-------|-------|-------|-------|
| ID          | ID | ID | ID | $X_0^3$ | $Y_0$ | $Z_0$ | $X_1$ | $Y_1$       | $Z_1$ | $X_2$ | $Y_2$ | $Z_2$ | $X_3$ | $Y_3$ | $Z_3$ |

Table 25: Measurement command byte 1 and 2

| CMD Byte #3 |          |          |          |          |          |     |           | CMD Byte #4 |  |  |  |  |  |  |  |
|-------------|----------|----------|----------|----------|----------|-----|-----------|-------------|--|--|--|--|--|--|--|
| $X_{02}^4$  | $Y_{02}$ | $Z_{02}$ | $X_{13}$ | $Y_{13}$ | $Z_{13}$ | VDD | $d^{(2)}$ | CRC[7:0]    |  |  |  |  |  |  |  |

Table 26: Measurement command byte 3 and 4

<sup>3</sup>  $X_0$  refers to Bx measurement of pixel 0

<sup>4</sup>  $X_{02}$  refers to differential Bx measurement between pixel 0 and pixel 2

## 15.4.2. Read Measurement command

A Read measurement command is commenced by sending 0x4 followed by an indication if the temperature is requested and the optional CRC if enabled.

| CMD Byte #1 |   |   |   |                | CMD Byte #2 |  |  |  |  |
|-------------|---|---|---|----------------|-------------|--|--|--|--|
| 0x4         | 0 | 0 | 0 | T <sup>5</sup> | CRC[7:0]    |  |  |  |  |

Table 27: Read measurement command

This command differs depending on the size and value for requested measurements. All measurements are given in the following order but only a maximum of 6 can be read out.

| CMD Byte #1 |    |    |    |                |                |                |                | CMD Byte #2    |                |                |                |                |                |                |                |
|-------------|----|----|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| ID          | ID | ID | ID | X <sub>0</sub> | Y <sub>0</sub> | Z <sub>0</sub> | X <sub>1</sub> | Y <sub>1</sub> | Z <sub>1</sub> | X <sub>2</sub> | Y <sub>2</sub> | Z <sub>2</sub> | X <sub>3</sub> | Y <sub>3</sub> | Z <sub>3</sub> |

| CMD Byte #3     |                 |                 |                 |                 |                 |     |                  | CMD Byte #4 |  |  |  |  |  |  |  |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----|------------------|-------------|--|--|--|--|--|--|--|
| X <sub>02</sub> | Y <sub>02</sub> | Z <sub>02</sub> | X <sub>13</sub> | Y <sub>13</sub> | Z <sub>13</sub> | VDD | d <sup>(2)</sup> | CRC[7:0]    |  |  |  |  |  |  |  |

Table 28: Read measurement response order

## 15.4.3. Read and Write registers

A Read register is commenced by sending 0x5 followed by the register address and the optional CRC if enabled.

| CMD Byte #1 |  |                        |  |  | CMD Byte #2 |  |  |  |  | CMD Byte #3 |  |  |  |  |
|-------------|--|------------------------|--|--|-------------|--|--|--|--|-------------|--|--|--|--|
| 0x5         |  | d <sup>(2)</sup> d d d |  |  | REG[7:0]    |  |  |  |  | CRC[7:0]    |  |  |  |  |

Table 29: Read register command

A write register is commenced by sending 0x6 followed by the register address, the two bytes to write and the optional CRC if enabled.

| CMD Byte #1 |   | CMD Byte #2 |   | CMD Byte #3 |          | CMD Byte #4 |           | CMD Byte #5 |  |
|-------------|---|-------------|---|-------------|----------|-------------|-----------|-------------|--|
| 0x6         | d | d           | d | d           | REG[7:0] | DATA[7:0]   | DATA[7:0] | CRC[7:0]    |  |

Table 30: Write register command

Before one is able to write register the following commands needs to be send first:

- 1) Send a first write command 0x9952 to register 0x43
- 2) Send a second write command 0x84DE to register 0x44

<sup>5</sup>To include temperature measurement  $T = 1$

#### 15.4.4. Exit command

The Exit command is stopping the repeated measurements if any and is switching the device in either READY, IDLE or SLEEP mode

An exit command is commenced by sending 0x8 followed by the Target MODE bits and the optional CRC if enabled.

| CMD Byte #1                | CMD Byte #2 |
|----------------------------|-------------|
| 0x8      TARGET MODE [3:0] | CRC[7:0]    |

*Table 31: Exit command*

- Ready Mode: 0x0
- Idle Mode: 0x1
- Sleep mode: 0x2

#### 15.4.5. Memory Recall Store Command

A memory Recall command is loading the content of the NVRAM into the registers.

| CMD Byte #1 | CMD Byte #2 |
|-------------|-------------|
| 0xD0        | CRC[7:0]    |

*Table 32: Memory recall command*

A memory Store command is storing the content of the registers into the NVRAM.

| CMD Byte #1 | CMD Byte #2 |
|-------------|-------------|
| 0xE0        | CRC[7:0]    |

*Table 33: Memory Store command*

#### 15.4.6. Reset Command

The RT command is used by the master to actively reset the device.

| CMD Byte #1 | CMD Byte #2 |
|-------------|-------------|
| 0xF0        | CRC[7:0]    |

*Table 34: Reset command*

15.5. SPI Communication

The SPI protocol is of mode 3, which means that data is clocked on rising edge of the clock and the clock is at a high level in idle (CPOL=1, CPHA=1). MSB are sent first. CS must be high after command completion and whenever the IC is not communicated with. With CS low the IC will wait for communication to occur.

15.5.1. Command implementation

For the examples give, the below convention is used.



Figure 3: SPI convention

15.5.1.1. SB, SWOC, SM

All the commands follow the structure below. The reply from the MLX90396 is only the status byte plus additional CRC. The example below is for a start of a burst mode with  $B_z$  components and differential  $B_z$  being measured.

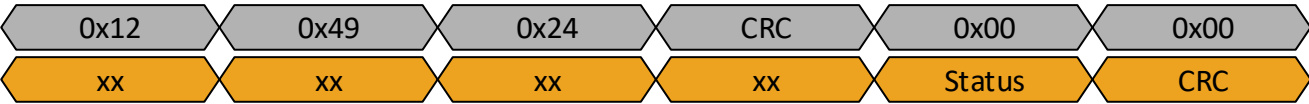


Figure 4: Example of a SB command,  $Z_0, Z_1, Z_2, Z_3, dbz02, dbz13$  in SPI

15.5.1.2. RM

Below are a few examples of a read measurement command.

For instance: a request was made for  $Z_0, Z_1, Z_2, Z_3, Z_{02}$  and  $Z_{13}$ .

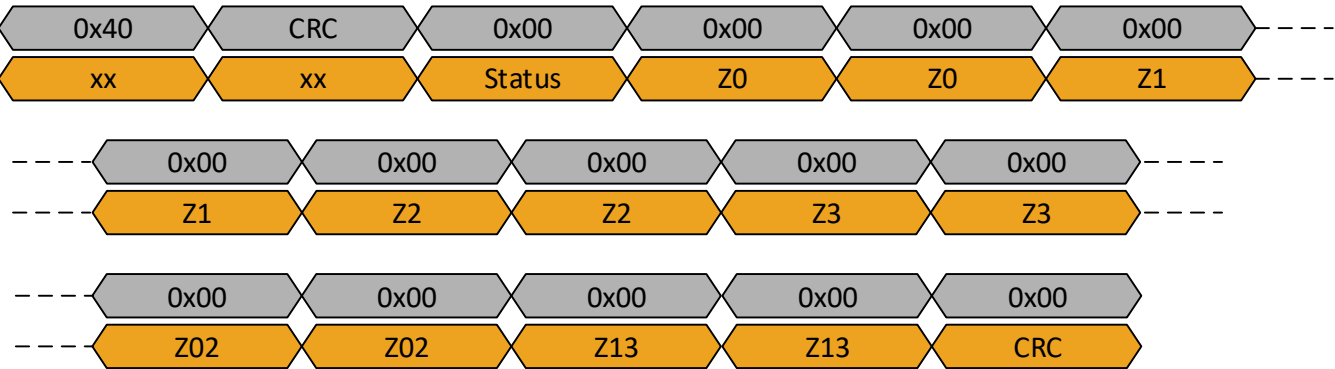


Figure 5: Example read measurement command in SPI

Another example for a XYZ measurement of pixel 0.

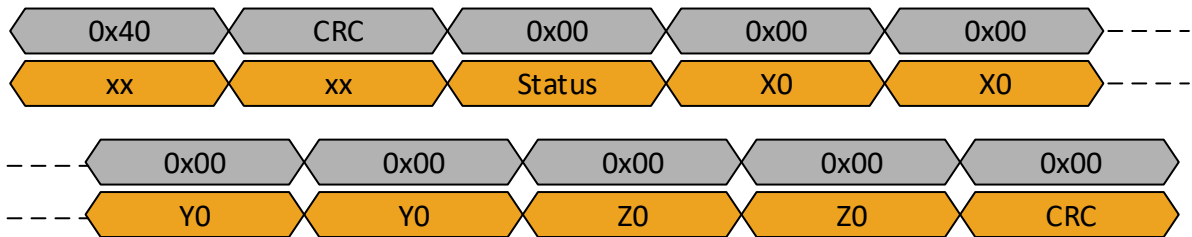


Figure 6: Example read measurement command in SPI

### 15.5.1.3. Read Register

An example of a Read register command.

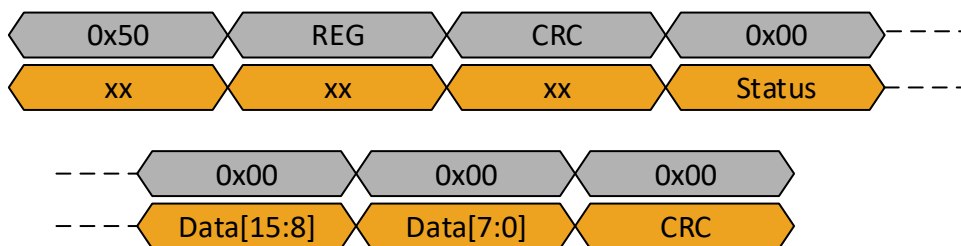


Figure 7: Example of a read register command in SPI

### 15.5.1.4. Write Register

An example of a Write register command.

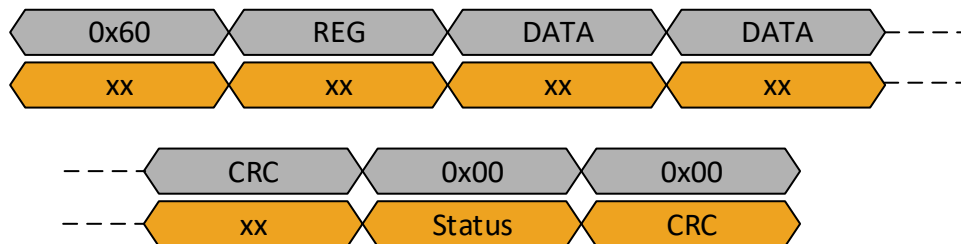


Figure 8: Example of a write register command in SPI

### 15.5.1.5. HS, HR, EX, RT

The structure of the RT, HS, HR and EX commands are the same, below an example of a HR command.

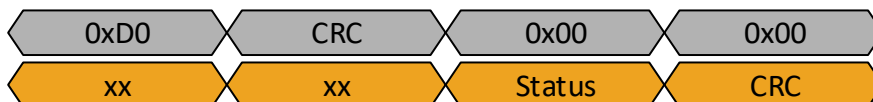


Figure 9: Example of a memory recall command in SPI

## 15.5.2. SPI Timing Specification

All timings are referred to the levels found in Table 5.

| Electrical Parameter    | Symbol         | Min. | Max. | Unit |
|-------------------------|----------------|------|------|------|
| SPI Clock Cycle         | $t_c$ (SPC)    | 100  |      | ns   |
| SPI Clock Cycle         | $t_c$ (SPC)    |      | 10   | MHz  |
| CS Setup Time           | $t_{su}$ (CS)  | 5    |      | ns   |
| CS Hold Time            | $t_h$ (CS)     | 10   |      | ns   |
| SDI Input Setup Time    | $t_{su}$ (SI)  | 5    |      | ns   |
| SDI Input Hold Time     | $t_h$ (SI)     | 15   |      | ns   |
| SDO Valid Output Time   | $t_v$ (SO)     |      | 50   | ns   |
| SDO Output Hold Time    | $t_h$ (SO)     | 5    |      | ns   |
| SDO Output Disable Time | $t_{dis}$ (SO) |      | 50   | ns   |

Table 35: General SPI Timing Specification

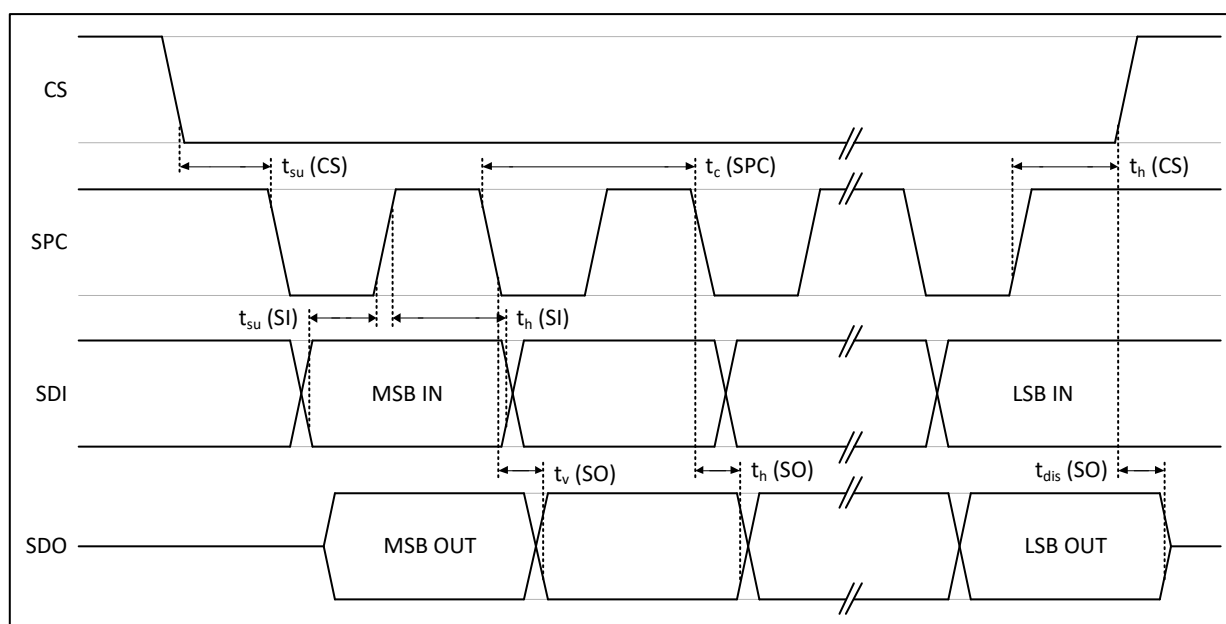


Figure 10: SPI Timing Diagram

## 15.6. I<sup>2</sup>C Communication

The I<sup>2</sup>C protocol is implemented such that the command is written to a specific register (0x80) in the RAM, and the status and measurement data are read back from this RAM. The commands themselves are the same as for SPI (section 15.4)

The address is determined by the MS\_A0\_A1 pin. Depending on the voltage that is supplied to that pin the address is selected.

- $0 < V(\text{MS\_A0\_A1}) < 1/8 \text{ VDD}$ : **SPI** mode
- $1/8 \text{ VDD} < V(\text{MS\_A0\_A1}) < 3/8 \text{ VDD}$ : **I2C** mode, **A0=0, A1=0**
- $3/8 \text{ VDD} < V(\text{MS\_A0\_A1}) < 5/8 \text{ VDD}$ : **I2C** mode, **A0=0, A1=1**
- $5/8 \text{ VDD} < V(\text{MS\_A0\_A1}) < 7/8 \text{ VDD}$ : **I2C** mode, **A0=1, A1=0**
- $7/8 \text{ VDD} < V(\text{MS\_A0\_A1}) < \text{VDD}$ : **I2C** mode, **A0=1, A1=1**

When the NVRAM parameter NV\_COMM\_SEL=0x2 the default I2C address shall be taken from NV\_COMM\_I2C\_ADDR which is 0x12.

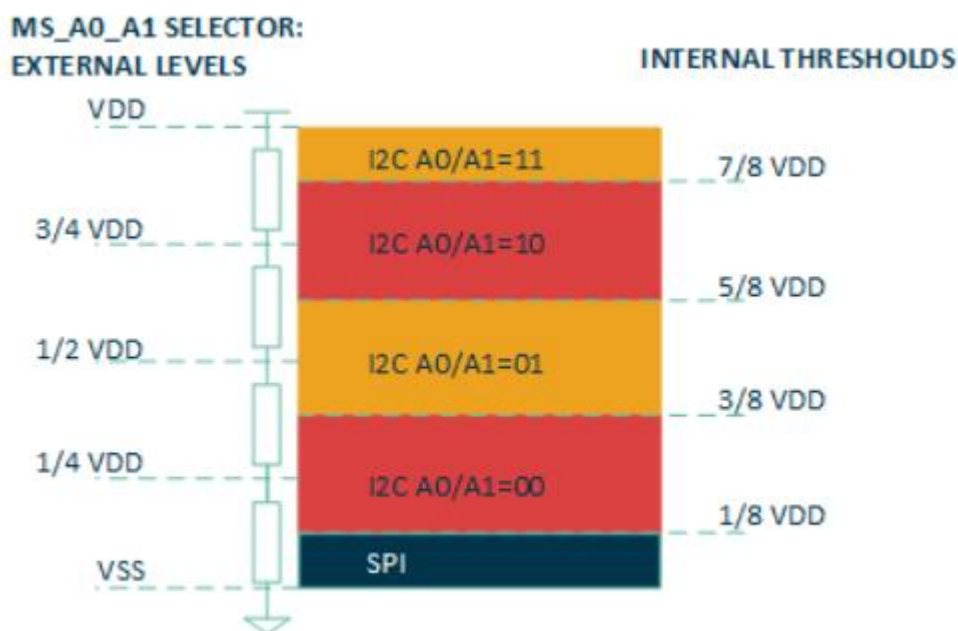


Figure 11: I2C hardware selection

### 15.6.1. Command Implementation

For the examples given, the below convention is used.

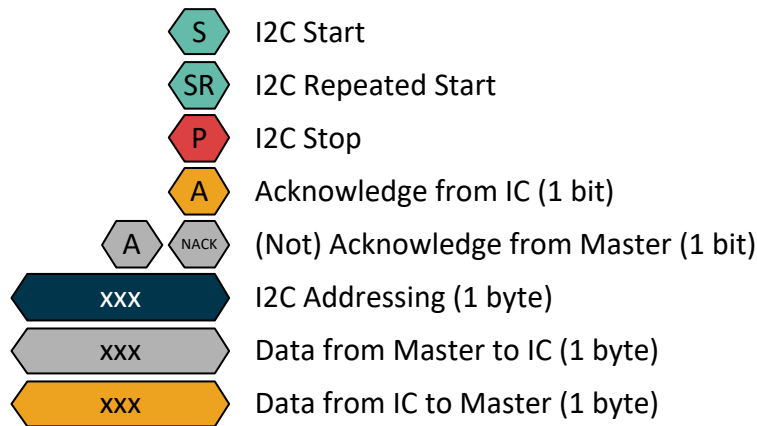


Figure 12: I<sup>2</sup>C Convention

#### 15.6.1.1. SB, SWOC, SM

All the commands follow the structure below. The reply from the MLX90396 is only the status byte. The example below is for a start of in burst mode for  $B_z$  components and differential  $B_z$  being measured.

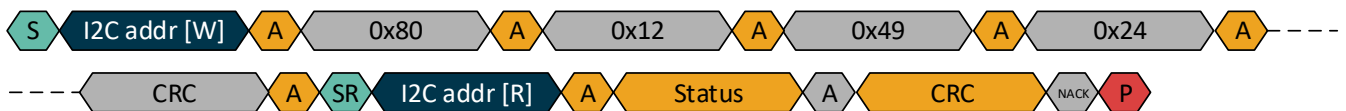


Figure 13: Example of a SB command in I2C

#### 15.6.1.2. RM

Below is an example of a read measurement where the requested data was the XYZ components of pixel 0.

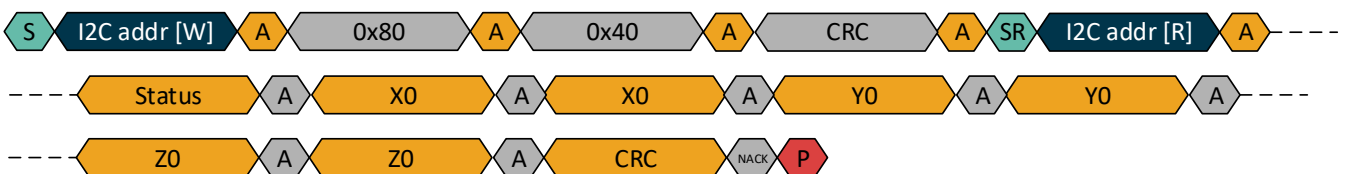


Figure 14: Example of a read measurement command in I2C

#### 15.6.1.3. Read Register

An example of a Read register command.

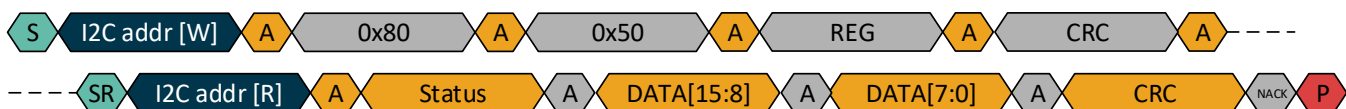


Figure 15: Example of a read register command in I2C

15.6.1.4. Write Register

An example of a write register command.

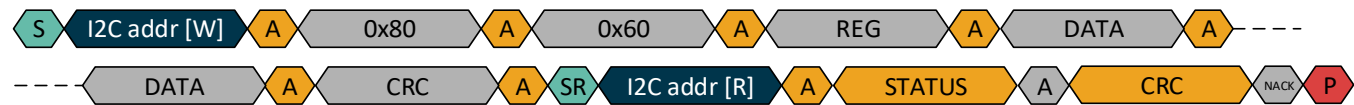


Figure 16: Example of a write register command in I2C

15.6.1.5. HS, HR, EX, RT

The structure of the RT, HS, HR and EX commands are the same, below an example of a RT command.

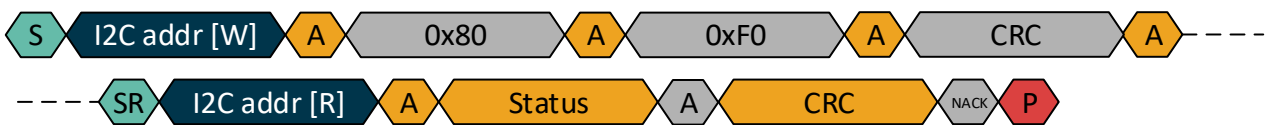


Figure 17: Example of RT command in I2C

## 15.6.2. I<sup>2</sup>C Timing Specification

All timings are referred to the levels found in Table 5.

| Electrical Parameter                           | Symbol                 | I <sup>2</sup> C Standard Mode |      | I <sup>2</sup> C Fast Mode Plus      |      | Unit |
|------------------------------------------------|------------------------|--------------------------------|------|--------------------------------------|------|------|
|                                                |                        | Min.                           | Max. | Min.                                 | Max. |      |
| SCL Clock Frequency                            | f (SCL)                |                                | 100  |                                      | 1000 | kHz  |
| SCL Clock Low Time                             | t <sub>w</sub> (SCLL)  | 4.7                            |      | 0.5                                  |      | μs   |
| SCL Clock High Time                            | t <sub>w</sub> (SCLH)  | 4                              |      | 0.26                                 |      | μs   |
| SDA Setup Time                                 | t <sub>su</sub> (SDA)  | 250                            |      | 50                                   |      | ns   |
| SDA Data Hold Time                             | t <sub>h</sub> (SDA)   |                                | 3.45 | 0                                    | 0.45 | μs   |
| SDA and SCL Rise Time                          | t <sub>r</sub> (SDA)   |                                | 1000 | 20+0.1*C <sub>b</sub> <sup>(6)</sup> | 120  | ns   |
|                                                | t <sub>r</sub> (SCL)   |                                |      |                                      |      |      |
| SDA and SCL Fall Time                          | t <sub>r</sub> (SDA)   |                                | 300  | 20+0.1*C <sub>b</sub> <sup>(6)</sup> | 120  | ns   |
|                                                | t <sub>r</sub> (SCL)   |                                |      |                                      |      |      |
| START Condition Hold Time                      | t <sub>h</sub> (ST)    | 4                              |      | 0.26                                 |      | μs   |
| REPEATED START Condition Setup Time            | t <sub>su</sub> (SR)   | 4.7                            |      | 0.26                                 |      | μs   |
| STOP Condition Setup Time                      | t <sub>su</sub> (SP)   | 4                              |      | 0.26                                 |      | μs   |
| Bus Free Time Between STOP and START Condition | t <sub>w</sub> (SP:ST) | 4.7                            |      | 0.5                                  |      | μs   |

Table 36: General I<sup>2</sup>C Timing Specification

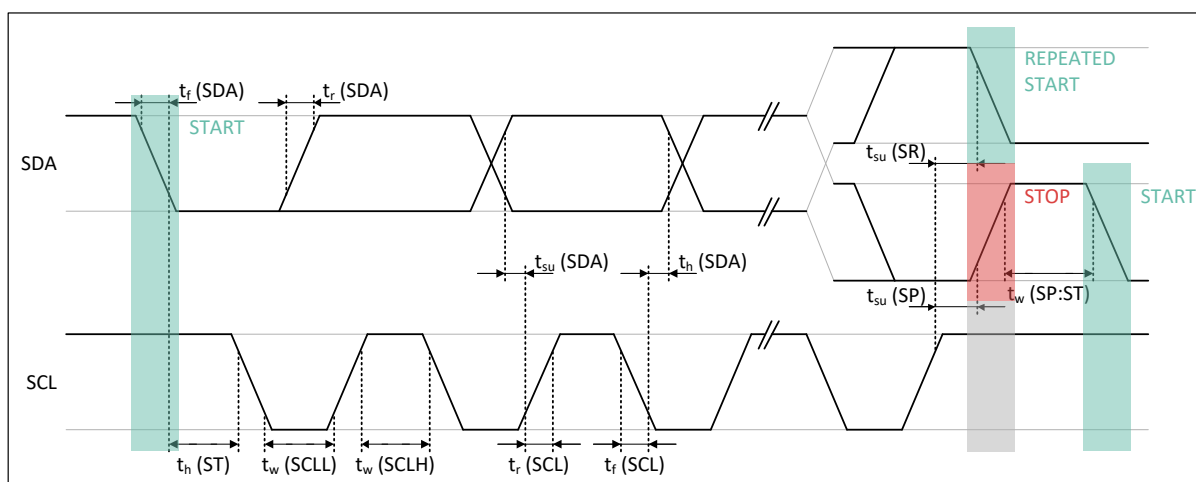


Figure 18: I<sup>2</sup>C Timing Diagram

<sup>6</sup> Where C<sub>b</sub> is the total bus capacitance (in pF).

## 15.7. CRC

By default, the MLX90396 shall check the CRC-8 byte of each incoming command when the CRC-8 protection is enabled and will report an invalid CRC-8 checksum in the info/Warn register. The CRC-8 protection can be disabled in the NVRAM.

The CRC is calculated with the following properties:

Polynomial:  $x^8 + x^5 + x^3 + x^2 + x + 1$

Compact notation: 0x2F

Koopmans notation: 0x97

Initial value: 0xFF (0x00 for AAA version)

Final XOR value: 0xFF (0x00 for AAA version)

A reset command in SPI mode the CRC results in 0x65



Figure 19: CRC example of a reset command in SPI

In case of a reset command in I2C mode the CRC is 0x82 with an I2C address = 0x13.



Figure 20: CRC example of a reset command in I2C

The CRC in I2C is calculated as follows:

As the CRC is calculated on the complete message including the I2C address taking into account the address was an I2C read or write.

- 1) Address 0x13 needs to be logically shifted  $0x13 \ll 1$  becomes 0x26.
- 2) Since this is a I2C write the LSB of the address is 0x0. 0x26 stays 0x26
  - a. In case of a read the LSB is 0x1. 0x26 becomes 0x27 in that case.
- 3) The CRC can now be calculated on [0x26 0x80 0xF0] resulting in 0x14.

## 16. Memory Structure and End-User Programmable Items

### 16.1. NVRAM Endurance

The NVRAM is qualified to guarantee an endurance of minimum 1000 write cycles (store operations) at 125°C (for engineering/calibration purpose).

If operated above 90°C with less than 3.0V supply, the performance of the NVRAM is not guaranteed.

### 16.2. Memory Structure

The MLX90396 has 1kbit of non-volatile memory, and the same amount of volatile memory. Each memory consists out of 64 addresses containing 16-bit words. The non-volatile memory has automatic 2-bit error detection and 1-bit error correction capabilities per address.

The memory is split in 2 areas:

- Customer area [address 0x00 to 0x27]
- Melexis area [address 0x28 to 0x31]

The RR and WR commands impact the volatile memory only, there no direct access possible to the non-volatile memory. The customer area of the volatile memory is fully accessible (read and write) to the customer; the Melexis area is write-protected and contains the 48-bit unique identification. Only modifications in the customer area are allowed with the WR command. The adjustments in the customer area can be stored in the permanent non-volatile memory with the store command HS, which copies the entire volatile memory to the non-volatile memory. With the HR command the non-volatile memory content can be recalled to the volatile memory, which can restore any modifications due to prior WR commands. The HR step is performed automatically at start-up of the IC, either through cold reset or warm reset with the RT command.

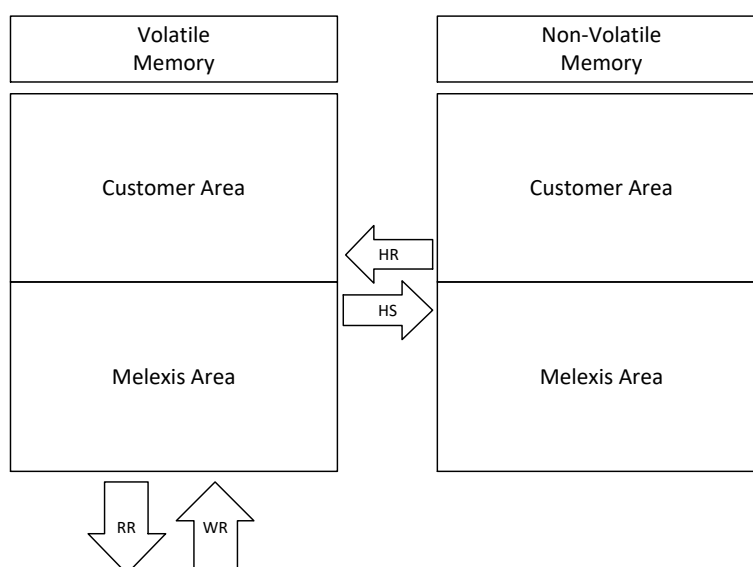


Figure 21: MLX90396 Memory Structure

### 16.3. End user programmable items

The table below gives an overview of the customer area in the MLX90396's NVRAM (BAA only).

| Parameter         | description                            | Addr. | offset | bits |
|-------------------|----------------------------------------|-------|--------|------|
| NV_GAINSEL_1PX    | gain single pixel measurement          | 0x00  | 0      | 6    |
| NV_GAINSEL_2PX    | gain dual pixel measurement            | 0x00  | 6      | 6    |
| NV_MAG_SCALE_Y    | Sensitivity X single-pixel measurement | 0x01  | 0      | 8    |
| NV_MAG_SCALE_Z    | Sensitivity Y single-pixel measurement | 0x01  | 8      | 8    |
| NV_MAG_SCALE_X02  | Sensitivity X dual-pixel measurement   | 0x02  | 0      | 8    |
| NV_MAG_SCALE_Y02  | Sensitivity Y dual-pixel measurement   | 0x02  | 8      | 8    |
| NV_MAG_SCALE_Z02  | Sensitivity Z dual-pixel measurement   | 0x03  | 0      | 8    |
| NV_MAG_SCALE_X13  | Sensitivity X dual-pixel measurement   | 0x03  | 8      | 8    |
| NV_MAG_SCALE_Y13  | Sensitivity Y dual-pixel measurement   | 0x04  | 0      | 8    |
| NV_MAG_SCALE_Z13  | Sensitivity Z dual-pixel measurement   | 0x04  | 8      | 8    |
| NV_MAG_K_TCL      | Thermal sensitivity correction (cold)  | 0x05  | 0      | 8    |
| NV_MAG_K_TCH      | Thermal sensitivity correction (hot)   | 0x05  | 8      | 8    |
| NV_MAG_K_TREF     | Thermal sensitivity correction ref     | 0x06  | 0      | 9    |
| NV_K_COMP_EN      | Sensitivity compensation enable        | 0x06  | 9      | 1    |
| NV_MAG_O_OREF_X0  | Offset X single-pixel measurement      | 0x07  | 0      | 8    |
| NV_MAG_O_OREF_Y0  | Offset Y single-pixel measurement      | 0x07  | 8      | 8    |
| NV_MAG_O_OREF_Z0  | Offset Z single-pixel measurement      | 0x08  | 0      | 8    |
| NV_MAG_O_OREF_X1  | Offset X single-pixel measurement      | 0x08  | 8      | 8    |
| NV_MAG_O_OREF_Y1  | Offset Y single-pixel measurement      | 0x10  | 0      | 8    |
| NV_MAG_O_OREF_Z1  | Offset Z single-pixel measurement      | 0x10  | 8      | 8    |
| NV_MAG_O_OREF_X02 | Offset X dual-pixel measurement        | 0x0A  | 0      | 8    |
| NV_MAG_O_OREF_Y02 | Offset Y dual-pixel measurement        | 0x0A  | 8      | 8    |
| NV_MAG_O_OREF_Z02 | Offset Z dual-pixel measurement        | 0x0B  | 0      | 8    |
| NV_MAG_O_OREF_X13 | Offset X dual-pixel measurement        | 0x0B  | 8      | 8    |
| NV_MAG_O_OREF_Y13 | Offset Y dual-pixel measurement        | 0x0C  | 0      | 8    |
| NV_MAG_O_OREF_Z13 | Offset Z dual-pixel measurement        | 0x0C  | 8      | 8    |
| NV_MAG_O_TREF     | Thermal offset ref temperature         | 0x0D  | 0      | 9    |
| NV_O_COMP_EN      | Offset drift enable                    | 0x0D  | 9      | 1    |
| NV_RANGE_O_TLCH   | Offset drift compensation range        | 0x0D  | 10     | 2    |
| NV_RANGE_O_OREF   | Offset compensation parameters         | 0x0D  | 12     | 2    |
| NV_MAG_O_TCL_X0   | Thermal offset correction (cold)       | 0x0E  | 0      | 8    |

|                     |                                      |      |    |    |
|---------------------|--------------------------------------|------|----|----|
| NV_MAG_O_TCH_X0     | Thermal offset correction (hot)      | 0x0E | 8  | 8  |
| NV_MAG_O_TCL_Y0     | Thermal offset correction (cold)     | 0x0F | 0  | 8  |
| NV_MAG_O_TCH_Y0     | Thermal offset correction (hot)      | 0x0F | 8  | 8  |
| NV_MAG_O_TCL_Z0     | Thermal offset correction (cold)     | 0x10 | 0  | 8  |
| NV_MAG_O_TCH_Z0     | Thermal offset correction (hot)      | 0x10 | 8  | 8  |
| NV_MAG_O_TCL_X1     | Thermal offset correction (cold)     | 0x11 | 0  | 8  |
| NV_MAG_O_TCH_X1     | Thermal offset correction (hot)      | 0x11 | 8  | 8  |
| NV_MAG_O_TCL_Y1     | Thermal offset correction (cold)     | 0x12 | 0  | 8  |
| NV_MAG_O_TCH_Y1     | Thermal offset correction (hot)      | 0x12 | 8  | 8  |
| NV_MAG_O_TCL_Z1     | Thermal offset correction (cold)     | 0x13 | 0  | 8  |
| NV_MAG_O_TCH_Z1     | Thermal offset correction (hot)      | 0x13 | 8  | 8  |
| NV_MAG_O_TCL_X02    | Thermal offset correction (cold)     | 0x14 | 0  | 8  |
| NV_MAG_O_TCH_X02    | Thermal offset correction (hot)      | 0x14 | 8  | 8  |
| NV_MAG_O_TCL_Y02    | Thermal offset correction (cold)     | 0x15 | 0  | 8  |
| NV_MAG_O_TCH_Y02    | Thermal offset correction (hot)      | 0x15 | 8  | 8  |
| NV_MAG_O_TCL_Z02    | Thermal offset correction (cold)     | 0x16 | 0  | 8  |
| NV_MAG_O_TCH_Z02    | Thermal offset correction (hot)      | 0x16 | 8  | 8  |
| NV_MAG_O_TCL_X13    | Thermal offset correction (cold)     | 0x17 | 0  | 8  |
| NV_MAG_O_TCH_X13    | Thermal offset correction (hot)      | 0x17 | 8  | 8  |
| NV_MAG_O_TCL_Y13    | Thermal offset correction (cold)     | 0x18 | 0  | 8  |
| NV_MAG_O_TCH_Y13    | Thermal offset correction (hot)      | 0x18 | 8  | 8  |
| NV_MAG_O_TCL_Z13    | Thermal offset correction (cold)     | 0x19 | 0  | 8  |
| NV_MAG_O_TCH_Z13    | Thermal offset correction (hot)      | 0x19 | 8  | 8  |
| NV_MAG_T_TCL        | $T_{sensor}$ slope correction (cold) | 0x1A | 0  | 8  |
| NV_MAG_T_TCH        | $T_{sensor}$ slope correction (hot)  | 0x1A | 8  | 8  |
| NV_MAG_T_TREF       | $T_{sensor}$ ref                     | 0x1B | 0  | 9  |
| NV_MAG_RO_XYZ01     | Rough offset correction Pixel0_1     | 0x1B | 9  | 7  |
| NV_MAG_RO_XYZ23     | Rough offset correction Pixel2_3     | 0x1C | 0  | 7  |
| NV_MAG_RO_XYZ0213   | Rough offset correction Pixel02_13   | 0x1C | 7  | 7  |
| NV_SUP_VDD_CORR     | Offset VDD measurement readout       | 0x1D | 0  | 8  |
| NV_COMM_I2C_ADDR    | I2C address                          | 0x1E | 0  | 7  |
| NV_PHY_MISO_CONFIG  | Miso configuration                   | 0x1E | 7  | 3  |
| NV_BURST_DATA_RATE  | Time between burst messages          | 0x1E | 10 | 6  |
| NV_WOC_MEAS1_THRESH | Threshold for measurement 1 (WOC)    | 0x1F | 0  | 12 |
| NV_ALGO_HP_COMP     | Diagnostic configuration             | 0x1F | 12 | 4  |

|                         |                                        |      |    |    |
|-------------------------|----------------------------------------|------|----|----|
| NV_WOC_MEAS2_THRESH     | Threshold for measurement 2 (WOC)      | 0x20 | 0  | 12 |
| NV_ALGO_HP_DEMOD        | Modulation configuration               | 0x20 | 12 | 4  |
| NV_WOC_MEAS3_THRESH     | Threshold for measurement 3 (WOC)      | 0x21 | 0  | 12 |
| NV_RETURN_TO_IDLE       | Return to Idle enable                  | 0x21 | 12 | 1  |
| NV_WOC_MODE             | WOC mode selection                     | 0x21 | 13 | 1  |
| NV_BURST_DATA_RATE_STEP | Burst delay resolution                 | 0x21 | 14 | 1  |
| NV_MEAS_COUNTER_MODE    | Measurement counter behavior           | 0x21 | 15 | 1  |
| NV_WOC_MEAS4_THRESH     | Threshold for measurement 4 (WOC)      | 0x22 | 0  | 12 |
| NV_RESET_INT_B          | Behavior INT_B                         | 0x22 | 12 | 1  |
| NV_SINGLE_TRIG_DIS      | Behavior CS_B                          | 0x22 | 13 | 1  |
| NV_COMM_CRC             | CRC enable                             | 0x22 | 14 | 1  |
| NV_PHY_RC_EN            | Rc filter enable                       | 0x22 | 15 | 1  |
| NV_WOC_MEAS5_THRESH     | Threshold for measurement 5 (WOC)      | 0x23 | 0  | 12 |
| NV_COMM_SEL             | Communication select (SPI/I2C)         | 0x23 | 12 | 2  |
| IA1_IN_SHORT_1PX        | Short inputs single pixel measurement  | 0x23 | 14 | 1  |
| IA1_IN_SHORT_2PX        | Short inputs dual pixel measurement    | 0x23 | 15 | 1  |
| NV_WOC_MEAS6_THRESH     | Threshold for measurement 6 (WOC)      | 0x24 | 0  | 12 |
| NV_WOC_TEMP_THRESH      | Threshold for Temperature (WOC)        | 0x25 | 0  | 12 |
| AFE_T_ADC_CNT_LSB       | ADC count configuration                | 0x25 | 12 | 2  |
| AFE_VDD_INPUT_MEAS      | Select VDD/INPUT measurement           | 0x25 | 14 | 1  |
| DIAG_UNDERTEMP_THRESH   | Threshold undertemperature diagnostic  | 0x26 | 0  | 4  |
| DIAG_OVERTEMP_THRESH    | Threshold overtemperature diagnostic   | 0x26 | 4  | 4  |
| NV_NVRAM_CRC_DIS        | Disable CRC NVRAM diagnostics          | 0x27 | 0  | 4  |
| NV_DIAG_ADC_DIS         | Disable ADC diagnostics                | 0x27 | 4  | 1  |
| NV_DIAG_HP_DIS          | Disable magnetic signal diagnostics    | 0x27 | 5  | 1  |
| NV_DIAG_IDLE_REF_DIS    | Disable idle_ref diagnostics           | 0x27 | 6  | 1  |
| NV_DIAG_INTB_DIS        | Disable int_B readout diagnostics      | 0x27 | 7  | 1  |
| NV_DIAG_LFO_DIS         | Disable LFO diagnostics                | 0x27 | 8  | 1  |
| NV_DIAG_OSC_DIS         | Disable main oscillator diagnostics    | 0x27 | 9  | 1  |
| NV_DIAG_RING_DIS        | Disable cracked ring diagnostics       | 0x27 | 10 | 1  |
| NV_DIAG_TEMP_DIS        | Disable temperature diagnostics        | 0x27 | 11 | 1  |
| NV_PHY_ANA_INT_DIS      | Disable analog fault reporting on INT  | 0x27 | 12 | 1  |
| NV_PHY_DIG_INT_DIS      | Disable digital fault reporting on INT | 0x27 | 13 | 1  |
| RESERVED                | Free space                             | 0x32 | 0  | 16 |
| RESERVED                | Free space                             | 0x33 | 0  | 16 |

|                    |            |      |   |    |
|--------------------|------------|------|---|----|
| RESERVED           | Free space | 0x34 | 0 | 16 |
| RESERVED           | Free space | 0x35 | 0 | 16 |
| RESERVED           | Free space | 0x36 | 0 | 16 |
| RESERVED           | Free space | 0x37 | 0 | 16 |
| NV_NVRAM_CRC16_MLX | NVRAM CRC  | 0x38 | 0 | 16 |

Table 37: End user programmable parameters

## 16.4. Melexis Area parameters (Read-Only)

| Parameter              | description | Addr. | offset | bits |
|------------------------|-------------|-------|--------|------|
| NV_HP23_XY_ROT_DIR     |             | 0x28  | 0      | 1    |
| NV_HP23_Z_ROT_DIR      |             | 0x28  | 1      | 1    |
| NV_HP01_XY_MODE_INIT   |             | 0x28  | 2      | 3    |
| NV_HP23_XY_MODE_INIT   |             | 0x28  | 5      | 3    |
| NV_HP01_Z_MODE_INIT    |             | 0x28  | 8      | 3    |
| NV_HP23_Z_MODE_INIT    |             | 0x28  | 11     | 3    |
| NV_AFE_T_AAF_BP        |             | 0x29  | 0      | 2    |
| NV_AFE_T_ST            |             | 0x29  | 2      | 1    |
| NV_AFE_T_ADC           |             | 0x29  | 3      | 1    |
| NV_AFE_T_ADC_CNT_MSB   |             | 0x29  | 4      | 1    |
| NV_AFE_T_SPIN_NB       |             | 0x29  | 5      | 2    |
| NV_ADC_DIAG_THRES      |             | 0x29  | 7      | 5    |
| NV_HP01P_XY_MODE_INIT  |             | 0x2A  | 0      | 3    |
| NV_HP23P_XY_MODE_INIT  |             | 0x2A  | 3      | 3    |
| NV_HP01P_Z_MODE_INIT   |             | 0x2A  | 6      | 3    |
| NV_HP23P_Z_MODE_INIT   |             | 0x2A  | 9      | 3    |
| NV_DIAG_HP_COMP_THRESH |             | 0x2B  | 0      | 12   |
| NV_ADC_DIAG_RUN        |             | 0x2B  | 12     | 3    |
| NV_TRIM_SUP_HFO        |             | 0x2C  | 0      | 7    |
| NV_TRIM_SUP_LFO        |             | 0x2C  | 7      | 5    |
| NV_TRIM_SUP_BIAS       |             | 0x2D  | 0      | 6    |
| NV_TRIM_SUP_BG         |             | 0x2D  | 6      | 5    |
| NV_TRIM_SUP_DIGREG     |             | 0x2D  | 11     | 5    |
| NV_NVM_REVISION_MAJOR  |             | 0x2E  | 0      | 4    |
| NV_NVM_REVISION_MINOR  |             | 0x2E  | 4      | 4    |
| NV_PRODUCT_CODE        |             | 0x2E  | 8      | 6    |
| NV_TEST_STATUS         |             | 0x2E  | 14     | 1    |

Table 38: Melexis Area Parameters

## 16.5. Melexis traceability parameters (Read-Only)

| Parameter         | description | Address | offset | bits |
|-------------------|-------------|---------|--------|------|
| NV_D_XPOS_ID      |             | 0x2F    | 0      | 8    |
| NV_D_YPOS_ID      |             | 0x2F    | 8      | 8    |
| NV_D_WFR_ID       |             | 0x30    | 0      | 5    |
| NV_D_LOT_ID_LSB11 |             | 0x30    | 5      | 11   |
| NV_D_LOT_ID_MSB6  |             | 0x31    | 0      | 6    |
| NV_D_FAB_ID       |             | 0x31    | 6      | 4    |
| NV_D_CORDAT_ID    |             | 0x31    | 10     | 6    |

Table 39: Melexis traceability parameters

## 16.6. Description of End user programmable parameters

### 16.6.1. GAINSEL

**NV\_GAINSEL\_1PX** defines the gain for single pixel measurements.

**NV\_GAINSEL\_2PX** defines the gain for dual pixel measurements.

### 16.6.2. MAG\_SCALE

**NV\_MAG\_SCALE** parameter defines the sensitivity for a specific axis. In case of **NV\_MAG\_SCALE\_Y** it would define the sensitivity of the Y-axis for all single pixel measurements. In case of **NV\_MAG\_SCALE\_Z02** it defines the sensitivity for the Z-axis for a dual measurement on pixel 0 and 2.

### 16.6.3. Thermal Sensitivity Correction

**NV\_MAG\_K\_TCL** is correction slope for in the cold range while **NV\_MAG\_K\_TCH** is the correction slope for the hot range. The split between cold and hot is defined by **NV\_MAG\_K\_TREF** and with **NV\_K\_COMP\_EN** this feature can be disabled.

| NV_K_COMP_EN | State    |
|--------------|----------|
| 0x1          | enabled  |
| 0x0          | Disabled |

Table 40: NV\_K\_COMP\_EN state

**NV\_MAG\_K\_TREF** is by default set for 35°C and is 2 complement with 1LSB/deg.

The gain at cold can be calculated with:

$$\frac{\left(\frac{NV\_MAG\_K\_TCL}{64}\right)}{128}$$

### 16.6.4. Offset compensation

**NV\_MAG\_O\_OREF\_X0** Is referring to the offset for axis X for single pixel measurements. This applies for both pixel 0 and pixel 2.

**NV\_MAG\_O\_OREF\_Y1** Is referring to the offset for axis Y for single pixel measurements. This applies for both pixel 1 and pixel 3.

**NV\_MAG\_O\_OREF\_Z02** is referring to the offset for axis Z for dual pixel measurements. This applies for using pixel 0 and pixel 2.

All previous offset parameters use **NV\_MAG\_O\_TREF** as reference temperature and can be disabled or enabled by **NV\_O\_COMP\_EN**.

| NV_O_COMP_EN | State    |
|--------------|----------|
| 0x1          | enabled  |
| 0x0          | Disabled |

Table 41: NV\_O\_COMP\_EN state

**NV\_MAG\_O\_TREF** is by default set for 35°C and is 2 complement with 1LSB/deg.

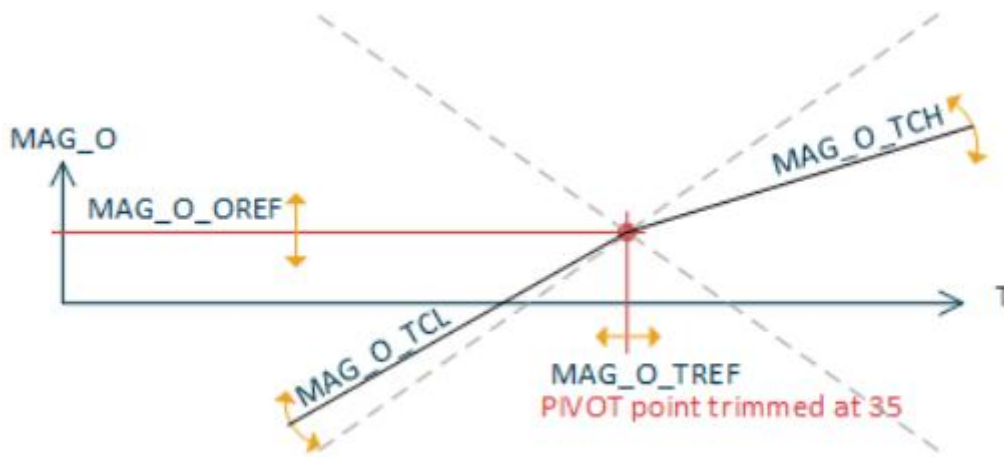


Figure 22: offset compensation

### 16.6.5. Temperature sensor

**NV\_MAG\_T\_CL** and **NV\_MAG\_THL** refers to the slope of correction of the temperature sensor using **NV\_MAG\_TREF** as a reference at 35°C. The parameters have a resolution of 1°C/LSB.

**NV\_WOC\_TEMP\_THRESH** can be used to start a measurement when the threshold is reached.

### 16.6.6. Rough offset

Referring to **NV\_MAG\_RO\_XYZ\_01** which is the rough offset correction for single pixel measurements (pixel 0 and pixel 1) for the X-Y and Z-axis. **NV\_MAG\_RO\_XYZ\_23** is the same but for pixel 2 and pixel 3.

**NV\_MAG\_RO\_XYZ0213** is for the rough offset correction in dual pixel measurement, pixel 0-2 and pixel1-3

At last **NV\_SUP\_VDD\_CORR** is to change the offset on the VDD measurement.

### 16.6.7. Communication configuration

When the sensor is configured to use I2C it uses the 5 MSB of **NV\_COM\_I2C\_ADDR** as address, the 2 LSB of the address are defined by the hardware configuration of MS\_A0\_A1 pin see Chapter 0.

**NV\_PHY\_MISO\_CONFIG** defines the state of the MISO driver.

| 0bxxx | MISO driver state     |
|-------|-----------------------|
| 0b0xx | Open-drain            |
| 0b100 | Push-pull             |
| 0bx00 | HIGH-Z                |
| 0bx01 | For 3 MHz clock speed |
| 0bx10 | For 6 MHz clock speed |
| 0bx11 | For 9 MHz clock speed |

Table 42: MISO driver state

**NV\_BURST\_DATA\_RATE** sets the interval of burst mode measurements and is related to **NV\_BURST\_DATA\_RATE\_STEP**.

| NV_BURST_DATA_RATE_STEP | description                                                |
|-------------------------|------------------------------------------------------------|
| 0x1                     | 1 LSB is a step of 20ms, for a range between 0 and 1260ms  |
| 0x0                     | 1 LSB is a step of 0.5ms, for a range between 0 and 31.5ms |

Table 43: NV\_BURST\_DATA\_RATE\_STEP configuration

With **NV\_RESET\_INT** the behavior of the INT\_B pin can be set as:

| NV_RESET_INT | State                                                              |
|--------------|--------------------------------------------------------------------|
| 0x1          | INT_B is automatically released at the start of a new measurement. |
| 0x0          | INT_B is released after a read measurement command                 |

Table 44: NV\_RESET\_INT configuration

The CS\_B pin can be used as a trigger in single measurement mode when **NV\_SINGLE\_TRIG\_DIS**=0x1. Programming it to 0x0 disables this function.

With **NV\_COMM\_CRC** the CRC for communication can be disabled or enabled.

| NV_COMM_CRC | State    |
|-------------|----------|
| 0x1         | enabled  |
| 0x0         | disabled |

Table 45: NV\_COMM\_CRC configuration

**NV\_PHY\_RC\_EN** = 0x1 enables an RC filter when communication speed goes above 7MHz, programming it to 0x0 disables this feature.

**NV\_RETURN\_TO\_IDLE** returns the sensors state automatically from the READY state to the IDLE state

| NV_RETURN_TO_IDLE | State                |
|-------------------|----------------------|
| 0x1               | Returns to Idle mode |
| 0x0               | disabled             |

Table 46: NV\_RETURN\_TO\_IDLE configuration

**NV\_MEAS\_COUNTER** describes the measurement counter behavior.

| NV_MEAS_COUNTER | State                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------|
| 0x1             | Counter increments after each measurement and is reset after a RM command                              |
| 0x0             | Counter increments after each measurement counter is reset after a new start burst measurement command |

Table 47: NV\_MEAS\_COUNTER configuration

### 16.6.8. Wake up on change thresholds

The MLX90396 has 6 wake-up on change thresholds these are referring to the 6 measurements that are selected. They are configurable with **NV\_WOC\_MEAS1\_TRESH** parameters

Depending on **NV\_WOC\_MODE** either the wake up on change is triggered on an absolute measurement or a relative measurement compared to the previous measurement that was performed.

| NV_WOC_MODE | State                            |
|-------------|----------------------------------|
| 0x1         | Relative to previous measurement |
| 0x0         | Absolute                         |

Table 48: NV\_WOC\_MODE configuration

### 16.6.9. Diagnostics

**NV\_DIAG\_UNDERTEMP\_TRESH** is used for Under-Temperature diagnostics, the resolution is 8°C/LSB.

| NV_DIAG_UNDERTEMP_TRESH | T threshold [°C] |
|-------------------------|------------------|
| 0b0000                  | 0 °C             |
| 0b0001                  | -8 °C            |
| ...                     | ...              |
| 0b1111                  | -120 °C          |

Table 49: NV\_DIAG\_UNDERTEMP\_TRESH resolution table

**NV\_DIAG\_OVERTEMP\_TRESH** is then again related to the Over-Temperature diagnostics, with a resolution of 16°C/LSB.

| NV_DIAG_OVERTEMP_TRESH | T threshold [°C] |
|------------------------|------------------|
| 0b0000                 | 0 °C             |
| 0b0001                 | 16 °C            |
| ...                    | ....             |
| 0b1111                 | 240 °C           |

*Table 50: NV\_DIAG\_OVERTEMP\_TRESH resolution table*

#### 16.6.9.1. Diagnostics

Each of the following diagnostics can be disabled or enabled. To disable a diagnostic, it has to be programmed to 0x1. With the exception of **NV\_NVRAM\_CRC\_DIS** to disable, it shall be programmed to 0xD.

| Diagnostic           | comment                                         |
|----------------------|-------------------------------------------------|
| NV_NVRAM_CRC_DIS     | Disables the NVRAM CRC check                    |
| NV_DIAG_ADC_DIS      | Disables the ADC diagnostic                     |
| NV_DIAG_HP_DIS       | Disable magnetic signal conditioning diagnostic |
| NV_DIAG_IDLE_REF_DIS | Disable IDLE_REF diagnostic                     |
| NV_DIAG_INTB_DIS     | Disable INTB readout diagnostic                 |
| NV_DIAG_LFO_DIS      | Disable Low-Frequency Oscillator diagnostic     |
| NV_DIAG_OSC_DIS      | Disable Main Oscillator diagnostic              |
| NV_DIAG_RING_DIS     | Disable Crack ring diagnostic                   |
| NV_DIAG_TEMP_DIS     | Disable over/under-temperature diagnostic       |
| NV_PHY_ANA_INT_DIS   | Disable analog fault reporting on INT_B pin     |
| NV_PHY_DIG_INT_DIS   | Disable digital fault reporting on INT_B pin    |

*Table 51: Diagnostic table*



## 17.2. Wiring with the MLX90396 in I2C mode

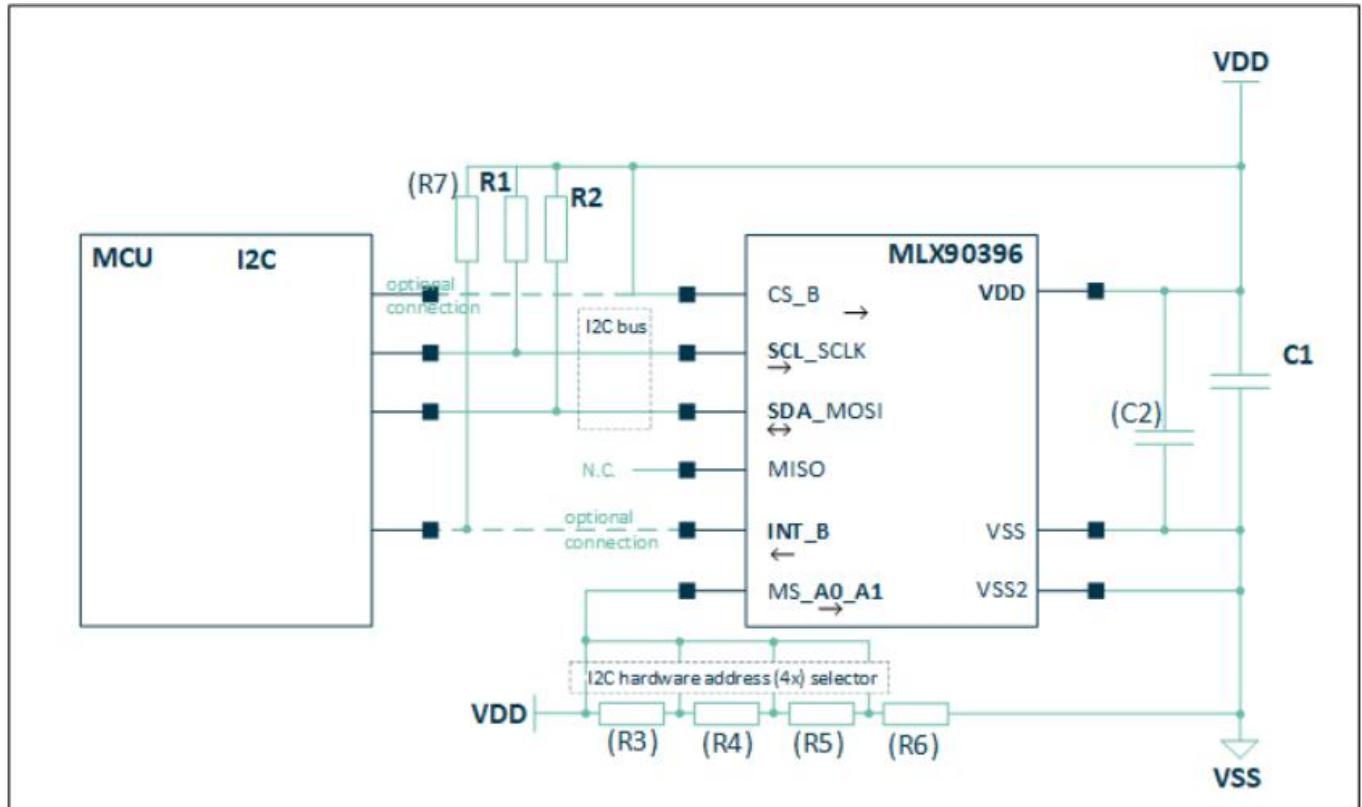


Figure 24: MLX90396 application diagram in I2C mode

| component          | Min.  | Typ.  | Max. | remark                                                                           |
|--------------------|-------|-------|------|----------------------------------------------------------------------------------|
| C1                 | 100nF |       | 1μF  |                                                                                  |
| C2                 |       | 100pF |      | Optimal, EMC robust routing                                                      |
| R1,R2 <sup>8</sup> |       | 10kΩ  |      | On MCU side, if not internal                                                     |
| R3,R4,R5,R6        |       | 100kΩ |      | Optional, I2C address selector<br>If not used MS_A0_A1 shall be connected to VDD |
| R7 <sup>9</sup>    | 80kΩ  | 100kΩ |      | Required for INT_B pin functionality                                             |

Table 53: Recommended values for QFN-16 in I2C mode

<sup>8</sup> To reach 1MHz in I2C R1 and R2 need to be less than 10kΩ.

<sup>9</sup> Has to as close as possible to the MCU

## 18. Standard information regarding manufacturability of Melexis products with different soldering processes

Our products are classified and qualified regarding soldering technology, solderability and moisture sensitivity level according to standards in place in Semiconductor industry.

For further details about test method references and for compliance verification of selected soldering method for product integration, Melexis recommends reviewing on our web site the General Guidelines soldering recommendation (<http://www.melexis.com/en/quality-environment/soldering>)

For all soldering technologies deviating from the one mentioned in above document (regarding peak temperature, temperature gradient, temperature profile etc.), additional classification and qualification tests have to be agreed upon with Melexis.

For package technology embedding trim and form post-delivery capability, Melexis recommends consulting the dedicated trim&form recommendation application note: lead trimming and forming recommendations (<http://www.melexis.com/en/documents/documentation/application-notes/lead-trimming-and-forming-recommendations>).

Melexis is contributing to global environmental conservation by promoting lead free solutions. For more information on qualifications of RoHS compliant products (RoHS = European directive on the Restriction Of the use of certain Hazardous Substances) please visit the quality page on our website: <http://www.melexis.com/en/quality-environment>.

## 19. ESD Precautions

Electronic semiconductor products are sensitive to Electro Static Discharge (ESD). Always observe Electro Static Discharge control procedures whenever handling semiconductor products.

20. Package information

20.1. QFN-16 package

20.1.1. QFN-16 package dimensions and sensitive spots location

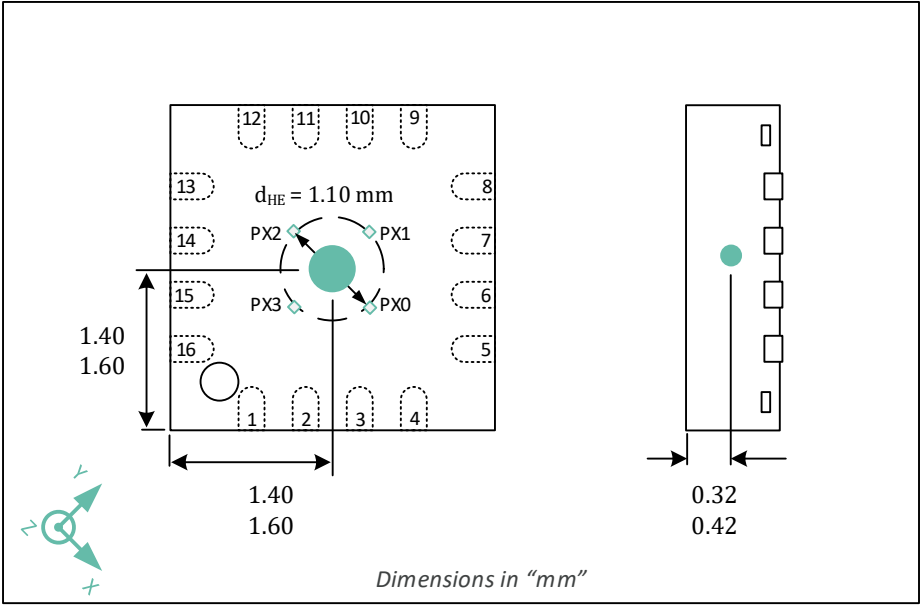
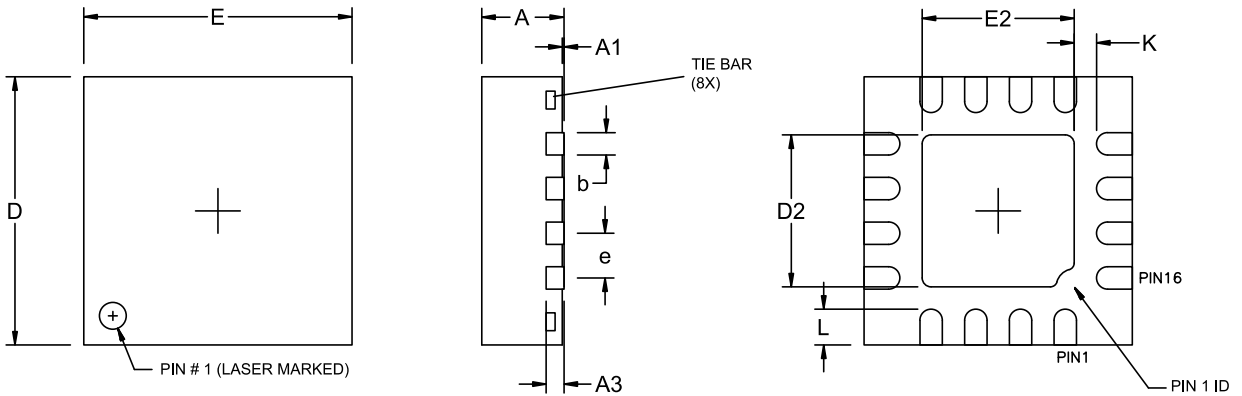


Figure 25: QFN-16 sensitive spot locations



NOTE :  
1. EXPOSED TIE BAR SHOULD BE KEPT FREE FROM SOLDER.  
2. (OPTIONAL) SIDE WALL IMMERSION TIN PLATING MIN 1um THICK.

| SYMBOL | ALL DIMENSION ARE IN MILLIMETERS |         |         |
|--------|----------------------------------|---------|---------|
|        | MINIMUM                          | NOMINAL | MAXIMUM |
| A      | 0.80                             | 0.85    | 1.00    |
| A1     | 0                                | 0.02    | 0.05    |
| A3     | 0.20 REF                         |         |         |
| D      | 2.90                             | 3.00    | 3.10    |
| E      | 2.90                             | 3.00    | 3.10    |
| D2     | 1.55                             | 1.70    | 1.80    |
| E2     | 1.55                             | 1.70    | 1.80    |
| L      | 0.30                             | 0.40    | 0.50    |
| K      | 0.15                             | ---     | ---     |
| b      | 0.18                             | 0.25    | 0.30    |
| e      | 0.50 BSC                         |         |         |

Figure 26: QFN-16 package dimensions

20.1.2. QFN-16 package pinout and marking

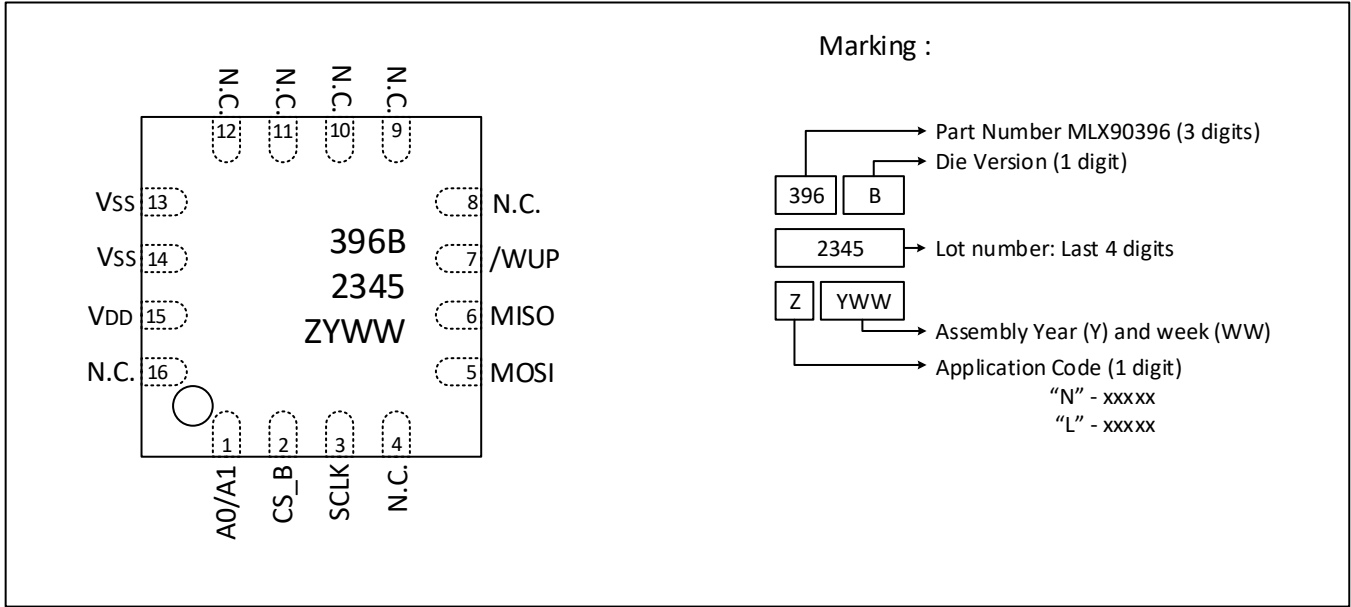


Figure 27: QFN-16 pinout and marking

## 21. Contact

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For additional information, please contact our Direct Sales team and get help for your specific needs:

|                |                                                                               |
|----------------|-------------------------------------------------------------------------------|
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